



An enhanced framework for real-time dense crowd abnormal behavior detection using YOLOv8

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Accepted: 24 March 2025 / Published online: 11 April 2025
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Abstract

Abnormal behavior detection in dense crowd, during the Hajj pilgrimage is vital to public security. Existing approaches face challenges due to factors like occlusions, illumination variations, and uniform attire. This research introduces the Crowd Anomaly Detection Framework (CADF), an improved YOLOv8-based model, integrating Soft-NMS to improve detection accuracy under complex conditions. CADF extensively evaluated on the Hajjv2 dataset, delivering an AUC of 88.27%, a 13.09% improvement over YOLOv2 and 12.19% over YOLOv5, with an Accuracy of 91.6%. To validate its generalizability, the framework is also tested on UCSD and ShanghaiTech datasets. Comparisons with state-of-the-art models, including VGG19 and EfficientDet, demonstrated CADF's superiority in accuracy, AUC, precision, recall, and mAP metrics. By addressing the unique challenges of Hajj crowd and achieving strong performance across diverse datasets, CADF highlights its potential for real-time crowd anomaly detection, contributing to enhanced safety in large-scale public gatherings and aligning with Sustainable Development Goals 3 and 11.

Keywords Abnormal behavior detection · YOLOv8 · Soft-NMS · Object detection · Crowd surveillance · Hajjv2 · Deep learning

1 Introduction

Advances in surveillance systems have led to an increase in the number of cameras in public areas, which has increased the significance of computer vision research (Alhothali et al. 2023). A CCTV camera positioned at various locations records surveillance videos, which are crucial for decision-making in detecting abnormalities, unlike GPS, mobile devices, or audio data, visual information provides more detailed insights, making it indispensable for monitoring and identifying anomalies. This interest has motivated researchers to explore

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methods for identifying and detecting anomalous events in video scenes to reduce reliance on the lengthy and costly procedure of manual inspection.

Anomaly detection methods face significant challenges in large, dynamic environments like the Hajj pilgrimage (Alaif et al. 2022a). Issues such as occlusion, where individuals are obscured by others, often result in missed detections and reduced accuracy. Furthermore, variations in illumination and the dynamic nature of crowd movements further complicate the accurate identification of anomalies (Albattah et al. 2020). The Hajj presents unique safety challenges characterized by dense crowd, uniform attire, and privacy concerns with millions of pilgrims gathering for a spiritual journey every year as shown in Fig. 1.

Human Activity Recognition (HAR) systems identify and analyze human actions in real time. Originally based on body movement projections, HAR has evolved to recognize interactions between humans and objects, providing deeper insights into abnormal behavior. This is particularly useful for crowd surveillance, where detecting suspicious or violent actions enhances public safety (Lentzas and Vrakas 2020). With rising violent incidents in urban and crowded spaces, automated violence detection in video streams is crucial for public safety. While CCTV is widely used, human monitoring alone is insufficient. AI-driven models, such as ResDLCNN-GRU, improve real-time recognition of violent behaviors, aiding timely interventions (Dey et al. 2024). Smart city surveillance further strengthens crime prevention by leveraging continuous AI analysis of video streams. The ability to detect physical altercations or crowd disturbances promptly is essential for maintaining public order (Ren et al. 2024), aligning with our study's goal of real-time anomaly detection during large-scale events like Hajj.

Current object detection algorithms struggle to differentiate individuals based on clothing variations, leading to frequent false positives and limited effectiveness in dynamic scenarios (Varghese and Thampi 2023). These shortcomings increase the risks of accidents and security breaches, underscoring the critical need for a robust anomaly detection system specifically tailored to address these challenges (Amosa et al. 2023). Detecting and responding

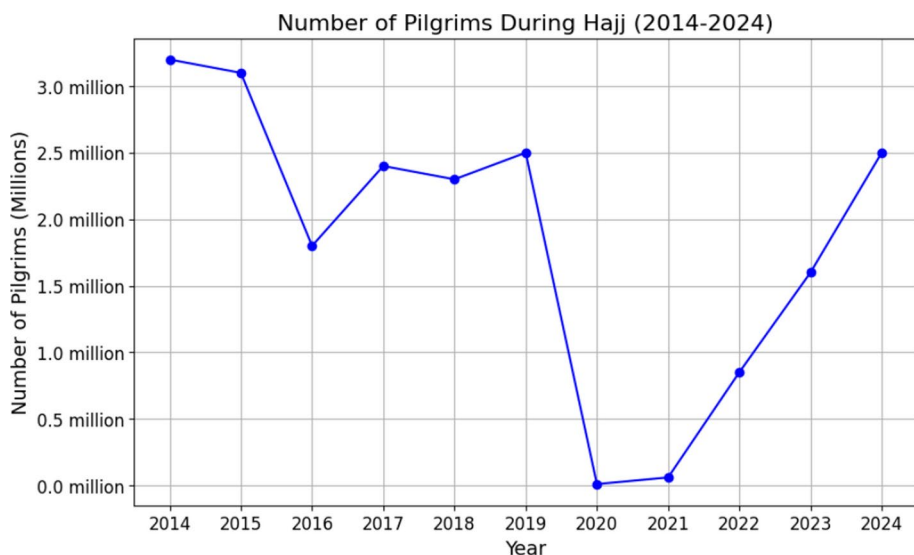


Fig. 1 Pilgrims perform religious activities during the Hajj

to behaviors such as erratic movements or disruptions in crowd flow are crucial to enhancing safety and security during the Hajj pilgrimage (Binsawad and Albahar 2022).

Computer vision is the leading area that focuses on object detection including identification and classification of objects in pictures or videos across a wide range of applications such as robotics, autonomous vehicles, and augmented reality (Deng et al. 2020), (Martinez-Martin and Pobil 2017). Initially, traditional methods such as support vector machines (SVM) were prevalent in this field but they had accuracy and scalability issues for today's use cases (Agrawal et al. 2020), (Alafif et al. 2023), (Felzenszwalb et al. 2010). The introduction of deep learning particularly deep convolutional neural networks (CNNs) has positively revolutionized object detection by greatly improving accuracy and robustness in complicated environments (Felzenszwalb et al. 2010), (Dalal and Triggs 2005). For computer vision abnormal behavior detection has become an essential discipline aiming at recognizing unexpected actions or events in different situations (Li et al. 2013). The definition of abnormal behavior varies contextually, influenced by factors such as crowd dynamics and environmental conditions. Detecting anomalies within crowded environments presents distinct challenges due to variations in crowd size, density, and behavioral patterns (Alafif et al. 2023). To prevent accidents, one must know when and where people will gather. This is because the organizers have to conduct detailed preliminary investigations as well to come up with a comprehensive plan for such public gatherings (Marsden et al. 2017). The management of crowd is effective through crowd analysis. Notably, there are usually prior notification given regarding popular human gatherings like religious events, sports events, carnivals or always crowded areas including airports, train stations and stadiums. Each gathering of people has some distinguishing features such as the aim, the venue, time and behavior of people there as well as their beliefs affiliations and race (Alafif et al. 2022a). Abnormal crowd behavior cause unusual events happenings in densely populated places which may suggest potential risks or threats to safety. Security can be maintained between various applications by recognizing aberrations in surveillance videos such as accident detection, crime detection, illegal activity detection, abandoned object detection and parking area monitoring (Hassner et al. 2012). Abnormal behavior identification is critical to saving people's lives during riots that may result into stampedes due to early interventions in crowded places. Researchers in computer science have focused a lot on this area and many algorithms and systems based on machine learning and computer vision for crowd detection and analysis have been developed. These systems are designed to identify patterns that deviate from the norm, enabling authorities to address potential security risks before they escalate into emergencies (Qasim and Bhatti 2019).

There are several main events that take place during Hajj including the stay at Mina tent city, circumambulating around Ka'aba which is known as Tawaf, and Rammi al-Jamarat (stoning of the devil). Besides, there is a stay at Muzdalifah and time spent at Arafat (Owaidah 2022). The Table 1 highlights significant incidents related to crowd management, disease outbreaks, and environmental factors that have impacted pilgrims during the Hajj (Wikipedia 2024).

The study focuses on critical issues in anomaly detection among densely populated areas, especially during large-scale events such as Hajj pilgrimage. Detecting abnormal behavior in dense crowd is a critical task in ensuring public safety. However, existing methods face persistent challenges, particularly in accurately identifying and detecting abnormal behaviors. These challenges often result in missed detections due to several factors. Occlusion in

Table 1 Historical incidents during Hajj

Date	Event	Death toll
24 September 2015	Rammi al-Jamarat	At least 2236
2 July 1990	Rammi al-Jamarat	1426
2024	Extreme Heat	At least 1301
12 January 2006	Rammi al-Jamarat	346
15 April 1997	Tent Fire	343
1 February 2004	Rammi al-Jamarat	251
23 May 1994	Rammi al-Jamarat	270
December 1975	Tent Fire	200
9 April 1998	Rammi al-Jamarat	118
11 September 2015	Crane Collapse	111
5 January 2006	Hotel Collapse	76
5 March 2001	Rammi al-Jamarat	35
11 February 2003	Rammi al-Jamarat	14
22 January 2005	Rammi al-Jamarat	3

**Fig. 2** Pilgrims perform religious activities during the Hajj

dense crowd obscures individuals, making it difficult for detection systems to consistently monitor and identify anomalies. Illumination variations, caused by fluctuating lighting conditions, further degrade the reliability of these models. In addition, uniform attire, common during events such as the Hajj, complicates the differentiation between individuals and their behaviors. The dynamic and unpredictable nature of crowd behavior adds another layer of complexity to real-time detection efforts. Existing approaches suffer from high false positives and false negatives, as they struggle to distinguish between normal and abnormal behaviors, leading to inconsistent results (Alhothali et al. 2023), (Alafif et al. 2023), (Alharthi et al. 2023), (Luo et al. 2023), (Alafif et al. 2022b), (Farooq et al. 2022), (Habib et al. 2021a). Addressing these challenges is essential for developing robust and reliable systems for the detection of abnormal behavior in dense crowd. That is why there is a need for a robust and effective framework specially developed to address these challenges, thereby enhancing safety and security for millions of participants. However, despite these advances, current algorithms tend to falter under such complexities unique to Hajj. Therefore, the aim of this research is twofold: first, it seeks to innovate on the currently used methodologies by incorporating state-of-the-art deep learning approaches capable of alleviating these issues; second, using the Hajjv2 dataset that contains real-world anomalies detected during the Hajj pilgrimage, it intends to rigorously evaluate these techniques. The objective here is therefore stated as addressing these specific issues will improve the accuracy, reliability, and applicability of abnormal behavior detection in highly populous dynamic environments. The research questions guiding this study include the following.

- **RQ1:** How can deep learning techniques be optimized to improve anomaly detection in dense crowd scenarios?
- **RQ2:** How can advanced deep learning techniques be integrated to address challenges such as occlusion and illumination variation?
- **RQ3:** What is their impact on the performance of anomaly detection systems in large-scale events like the Hajj pilgrimage?
- **RQ4:** How effectively does the CADF detect anomalies in complex and dynamic environments compared to other state-of-the-art models?
- **RQ5:** How does the generalizability of CADF improve anomaly detection across different crowd densities and environments?

By addressing these questions, this research aims to contribute novel insights to enhance the safety and security of the Hajj event by effectively identifying and classifying abnormal behaviors amidst dense crowd and dynamic environmental conditions.

To address these limitations effectively, the CADF is proposed in this study. This framework enhances You Only Look Once version 8 (YOLOv8) using a modular backbone and multi-scale feature fusion, crucial for large crowd sizes, with diverse behaviors of Hajj gatherings. In addition, Soft Non-Maximum Suppression (Soft-NMS) was applied to enhance anomaly detection by removing duplicate detections and improving spatial intelligence in obscured and congested scenes mostly seen during the Hajj period. The Hajjv2 dataset has been employed in our work, which comprehensively annotated bounding box coordinates and object classes for training purposes. We took frames from annotated videos, converted them into JPEG images, and made corresponding YOLO-formatted label files out of them. Training was done using pre-trained YOLOv8 weights with 20 epochs for optimal performance under Hajj-specific conditions. The model's effectiveness was evaluated based on AUC, precision, recall, F1-score, and mAP metrics indicators, which revealed significant improvements over YOLOv2 as well as YOLOv5 benchmarks. Additionally, to ensure the robustness and generalizability of CADF, we tested the model on the UCSD and ShanghaiTech datasets. Furthermore, we compared CADF with state-of-the-art object detection models, including VGG19 and EfficientDet, demonstrating superior performance in detecting anomalies in complex crowd scenarios. During large-scale events such as the Hajj pilgrimage, this methodology emphasizes CADF's efficacy in improving anomaly detection and ensuring public safety. This study aligns with SDGs 3 (Good Health and Well-being) and 11 (Sustainable Cities and Communities) by leveraging technological innovations to enhance safety and security measures in crowded urban areas. The CADF contributes to more effective crowd management and enhances public safety through better anomaly detection during events such as Hajj, thus supporting sustainable development initiatives. Several gaps are addressed by this study in abnormal behavior detection from surveillance footage, especially in situations characterized by high crowd density and complex dynamics. This research contributes to the field of abnormal behavior detection in the following key aspects:

- We propose a novel approach, CADF, by integrating **Soft-NMS into the YOLOv8 framework**. This enhancement improves the model's ability to detect objects under challenging conditions such as **occlusions and illumination variations**. The integration of Soft-NMS reduces missed detections while preserving crucial overlapping detec-

tions, leading to improved accuracy in real-world dense crowd scenarios.

- Our model is rigorously evaluated on multiple datasets, including the Hajjv2 dataset (capturing real-world anomalies from Hajj pilgrimage videos), as well as benchmark datasets **UCSD** and **ShanghaiTech**. This cross-dataset evaluation ensures the robustness and generalizability of CADF across diverse crowd environments, demonstrating its adaptability beyond religious gatherings.
- In addition to benchmarking CADF against baseline YOLO models (YOLOv2, YOLOv5), we conduct comparative experiments with state-of-the-art object detection models, including **VGG19** and **EfficientDet**. Our results show that CADF outperforms these models in terms of **AUC**, **precision**, **recall**, **F1-score**, and **mean Average Precision (mAP)**, establishing it as a highly competitive approach for real-time abnormal behavior detection.
- Our research provides practical solutions for **real-time crowd monitoring and safety management**, especially in large-scale public events. The early detection of abnormal behavior enables proactive intervention, reducing security risks such as stampedes, crowd surges, and disorderly movements. The deployment of CADF in **smart surveillance systems** can significantly enhance public safety measures in high-density gatherings.
- By improving anomaly detection in crowded environments, this research supports **SDG 3** and **SDG 11**. The ability to monitor and manage public spaces effectively through AI-driven surveillance contributes to safer and more resilient urban environments.

The rest of the paper is organized as follows: we provide a brief background of study in Sect. 2. A description of literature related to abnormal behavior detection and recognition is presented in 3. This is followed by an overview of the abnormal behavior Hajjv2 dataset and your intended approach in 4. Procedures for implementing the experiments, results and data analysis are then presented in 5. Discussion on the evaluative experiments carried out, the challenges and limitations of the study are presented in Sect. 6. As a conclusion, 7 summarizes the work done and a section of recommendations for further searches is presented at the end of the paper.

2 Background

A lot of research has been conducted in abnormality detection domain due to its importance and complexity in computer vision field for last two years (Alaif et al. 2023), (Duong et al. 2023), (Cui et al. 2023), (Jha and Babiceanu 2023), (Mahadevkar et al. 2022), (Tran et al. 2022). The video anomaly detection and localization largely depends on two things such as environment complexity and types of the anomalies. Additionally, environment complexity is affected by the number of moving targets that are present in a scene. Hence, based on the density of the moving targets, the environment can be classified into three types such as sparsely crowded environment (loose crowd, i.e., 10 sqft/person), moderately crowded environment (more dense crowd, i.e., 4.5 sqft/person), densely crowded environment (very dense crowd, i.e., 2.5 sqft/person).

Hajj is a yearly religious gathering for Muslims. Annually, on every pilgrimage millions of people from any region gather, regardless of their age or color and cultural development

engage into a pilgrimage to Saudi Arabia, more exclusively the places like Mecca, Medina to observe Hajj and Umrah (Albattah et al. 2020) as illustrated in the figure. This kind of diversity in crowd would not have occurred in the absence of certain recent technological advancements which include wireless networking, computer vision or spatial computing, data analytics and mobile applications among other (Felemban et al. 2020). Summoning the events conducted for participants' safety, mainly factors such as crowd monitoring and human activity detection as defining risks are glasses, watch scanners whose systems have such technologies as looking at crowd behavior strategy (N. Ibadah and Oumsis 2020). It is important to note, that despite the considerable efforts of the Saudi authorities in the monitoring of more than one hundred thousand pilgrims using several thousands of wireless visual sensor networks and several scalar sensor technologies mobile phone GPS sensor for tracking and monitoring pilgrims, monitoring of such event becomes a very difficult task because of high number of pilgrims who are present for the event each year (Habib et al. 2021b).

The Hajj journey is a multi-day religious observance, which brings together millions of Muslim persons annually in Mecca starting on the 8th of Dhu al-Hijjah known as the Day of Tarwiyah (Yiğit 2021). These pilgrims move to Mina where they spend night in prayer. Narrow streets of Mina become tightly packed at this stage resulting in abnormal behaviors such as people just standing or sitting on walkways to block the flow of pedestrians (Al Ruwaithi 2021a). Furthermore, complicating movement and increasing the risk of occlusion, where certain areas become impassable due to overcrowding (Namoun et al. 2018). On 9th day called Day of Arafat, the pilgrim assemble in plains calling it a day were they stand still praying which marks important part in Hajj rituals (Peters 2021). Managing crowd becomes very challenging during this big gathering with certain people moving opposite from the direction of majority leading into confusion and safety risks. As sun sets there might be lighting problems that may hinder surveillance systems from detecting these unusual acts such like running or engaging in non-pedestrian activities like operating cars or wheelchairs in inappropriate areas (Noorwali et al. 2024). On the 10th day of pilgrimage, known as Day of Eid pilgrims are to return to Mina and stone Jamarat. This ritual involves throw pebbles at pillars, symbolizing rejecting evil spirits. The intensity of this activity coupled with huge numbers of people can result in unusual running and pushing that eventually develops into dangerous circumstances, for example, stampedes. However, there are more people moving in different directions hence increasing the risk of collisions and injuries in Massaa (Owaidah 2015). The 11th and 12th days mark additional stoning rituals at Mina by the pilgrims together with Tawaf performed within Mecca which involves circumambulating around the Kaaba. Some common abnormal activities that are often seen in these crowded places where people worship include standing or sitting on unauthorized spots like statues or trees; sleeping in front of crowd; using vehicles such as cars and wheelchairs among others which are not supposed to be used within restricted areas meant for pedestrians only (Al Ruwaithi 2021b). Furthermore, because many muslims wear white during Tawaf, it may be difficult to see exactly who is acting out inappropriately during these rituals without proper monitoring systems that ensure everyone's safety even when they're doing something wrong (Halboob et al. 2024).

Hajj is a complex and dynamic environment where safety and logistical operations are seriously affected when there is an abnormal behavior. Unusual and potentially dangerous activities such as crowd surges, stampedes, or some individuals going against usual paths

are possible because of the huge size and compactness of the crowd (Alharbey et al. 2022). Sudden changes in movement patterns, excessive clustering or erratic actions can interrupt flow of the pilgrims posing serious risks. Such behaviors may arise from different reasons including panic, confusion or even physical stress caused by long distance journey. For this reason, identifying and managing these abnormalities is essential for safety and well-being of all participants thus necessitating development of advanced detection and monitoring systems. Proper handling of abnormal behavior helps to prevent accidents while at the same time enhancing control over crowd hence making the pilgrimage safer and more organized (Al-Nabhan et al. 2021). One must compare the present ways of handling abnormal behaviors during the Hajj with those applied in the past. Such contrasts will demonstrate how new strategies and technologies have improved crowd safety and movement. By examining previous occurrences and control methods, one can be able to tell whether they were successful or failed thereby providing for more efficient alternatives in forthcoming pilgrimages. This constant assessment and adjustment program is crucially important to intensifying general security and productivity of Hajj rituals so as to serve ever changing requirements of millions who are coming each year (Abalkhail and Al Amri 2022).

A great number of pilgrims crowd at the same time in the Hajj area, which constitutes an excellent example of a high-density environment. In this type of environment, various abnormal activities may occur, such as blockages, walking against the flow of pilgrims, standing on pathways and obstructing movement, sitting in unauthorized areas, and rushing or storming gates and train stations (Ullah et al. 2019). Violence can also be described as any use of physical force upon others including people around or against them in general through a smart surveillance system that maintains safety reducing violations and other accidents (Saini et al. 2016). Human abnormal behavior may be grouped into three different forms as shown in Table 2 which include point anomalies where an individual's conduct differs from that of others (Basora et al. 2019). Collective anomalies detected among a set of related entities portraying patterns correlated to some other data (Canizo et al. 2019). Contextual anomalies also termed conditional ones where certain occurrences become unusual only under specific contexts (Shaukat et al. 2021).

Table 2 Type of anomaly and matching situations in Hajj

Type of anomaly	Matching situations in Hajj
Point anomalies	- People walking in the opposite direction to the flow of pilgrims - Individuals sitting in areas designated for walking - People sleeping in pathways meant for pilgrims
Collective anomalies	- Groups of people walking in the opposite direction to the main crowd flow - Groups engaging in fights or violent behavior
Contextual anomaly	- Sitting in areas other than those designated for pilgrims, which is considered abnormal during Hajj, despite being normal behavior in other public settings like parks

3 Related work

In recent years, the management of crowd during large-scale events such as the Hajj pilgrimage has become a critical area of research due to the potential risks associated with dense populations. Effective crowd management during Hajj involves strategies for monitoring and controlling the movement of millions of pilgrims to ensure safety and prevent incidents. Researchers have explored various techniques for crowd management to detect abnormal behaviors that could indicate potential safety hazards. Abnormal behavior detection during Hajj focuses on identifying behaviors that deviate from normal crowd patterns, such as sudden movements or congestion, which require immediate attention. This section explores the existing literature on crowd management and abnormal behavior detection during Hajj, highlighting the contributions of YOLOv8 based methods. It also identifies the key issues, which have been prevalent in existing methods. Understanding these issues helps to outline the research gaps and limitations, setting the stage for addressing these challenges through the proposed research.

3.1 Crowd management during Hajj

Hajj and Umrah events are particularly challenging and ideal activities to assess the effectiveness of crowd management systems because of the large scale and heterogeneity of the participants as well as the nature of the context. The Table 3 compares different crowd management techniques during Hajj, showing the methods used, the datasets involved, and the techniques applied in each study. It covers various approaches such as detecting abnormal behavior, tracking and localization of abnormal behavior, estimating crowd density, and counting people in crowd.

In the articles by Nafea (Nafea 2021) and Owaidah et al (Owaidah et al. 2021) estimated computational methods for flow regulation as well as for crowd control impressively.

Table 3 Comparative analysis of crowd management techniques

Study	Crowd management technique	Dataset	Techniques used
Alaif et al. (2023)	Abnormal behavior and tracking	Hajjv2	CNN, RF
Alharbey et al. (2022)	Localization and tracking	Dataset includes annotated faces images from internet	Yolov4
Bhuiyan et al. (2022)	Density estimation	Hajj dataset in the tawaf area. Source: Youtube	FCNN
Alaif et al. (2022b)	Abnormal behavior	Various pilgrimage videos demonstrating strange sensibilities	GAN, optical flow
Al-Ahmadi et al. (2021)	Crowd movement behaviors	Hajjv1	ABS
Bhuiyan et al. (2021)	Estimating density and counting crowd	YouTube crowd counting dataset for various Hajj locations and scenes	CNN
Habib et al. (2021c)	Crowd counting	In the crowd counting dataset Hajj that includes tawaf area. Source: Internet	CNN, Faster R-CNN
Nafea (2021)	Density estimation	Dataset of moving crowd	CNN, ABS
Al-Nabhan et al. (2021)	Localization and behavior monitoring	pilgrims' movement data: RFIDs; Environmental sensors	simulation based on GIS and IoT

Table 4 Comparative analysis of different techniques for abnormal behavior detection

Reference	Objective	Techniques	Dataset	Result	Limitations
Alaif et al. (2023)	Provide a thorough approach for identifying anomalous behavior in crowd, for both small-scale and large-scale crowd	YOLOv2, Kalman filter techniques	UMN, UCSD (Small-scale), Hajj1 and Hajj2 (Large-scale)	Achieved 76.08% average AUC on the Hajjv2 dataset and 99.76% and 93.71% average AUCs (small-scale)	Challenges with heavy, partial, and full occlusions. Need to enhance detection and recognition in large-scale crowd
Alharthi et al. (2023)	Extract appearance and motion information for enhanced accuracy in classifying abnormal behaviors within crowd	C3D and VGG-16	Hajj2	Precision, accuracy, and F1-score of C3D: 0.90, 0.89, 0.88, respectively	Needs improvement in robustness and generalizability
Alhothali et al. (2023)	Detect and locate anomalies in dense crowd scenes	Yolov5 (detection), Deep SORT (Tracking), SVM (classification)	Hajj2 Dataset	AUC of 88.96% overall, with AUCs of 95.30% for Masa'a, 65.60% for Jamarat, and 92.11% for Arafat	Computational challenges affecting real-time operation. Need for faster processing for detection
Luo et al. (2023)	Develop a VAD method for large-scale crowd-level abnormal behavior detection	Graph-based method, multi-scale motion consistency network (MSMC-Net)	UMN, Hajj, Love Parade	AUC UMN: 94.4, Hajj: 92.3	Computational complexity. Need for model enhancement to address illumination variation
Alaif et al. (2022b)	Detect abnormal behavior in the Hajj large-scale crowd dataset	Generative Adversarial Network (GAN), U-Net model, Transfer Learning	UMN, UCSD, Hajj	UMN scenes 1, 2, 3, and UCSD: 99.4%, 97.1%, 97.6%, 89.26% respectively, Hajj dataset: 79.63%	Complexity of detection, challenges in achieving high accuracy. Limitations in addressing occlusion and illumination variations
Farooq et al. (2022)	Developed a new method for detecting abnormal behavior in crowd, which could lead to disasters like stampedes	PILGRIM dataset	Accuracy	Good performance on dataset at low density and high density crowd	Inconsistent in abnormal detection in high density crowd
Habib et al. (2021a)	Develop framework to identify abnormal activities using lightweight CNN for pilgrims at Mecca	Lightweight CNN	Hockey fight, surveillance fight	Dataset1 achieved accuracy 98 and dataset 2 achieved 81.05	Model detects violent activity from a single perspective, limiting anomaly information to crowd boundaries, making global anomaly detection in high-density crowd challenging

Owaidah et al. (Owaidah et al. 2021) presented the figure of the oppositExceptions programmed which simulated crowd behavior using video feeds. Also, the framework application is described during the pilgrimage season at the holy site of Rawdah in the Great Mosque of Saudi Arabia, emphasizing the appropriate management of crowd concerning movement and space utilization. Furthermore, the simulation of large crowd during the Hajj pilgrimage in Mecca was described by Al Ahmadi et al. (Al-Ahmadi et al. 2021). A significant number of participants rotate around Al-Ka'ba from Safa and Marwa hills as part of the study's description of the integration of Tawaf and Sayee rituals. Bhuiyan et al. (Bhuiyan et al. 2021) resorted to modeling along with computer software Expand Sim to develop D.E.S and simulate movement of pilgrims during the rituals considering factors like crowd density, flow of people, speed of walking etc. The simulation results give courses of how much the behavior of the crowd would be altered according to the best provenients crowd management techniques with noticeable pedestrian flow effects.

Al-Nabhan et al. (Al-Nabhan et al. 2021) suggested a crowd management technique which employs the principles of Internet of Things. Such a technique deploys quite a number of sensors which include radio frequency identification (RFID), environmental sensors as well as cameras that collect crowd related information including specifications and positioning of the pilgrims, weather and moisture conditions, and images in real-time. The information collected is processed to detect areas that are likely to be dangerous and become over-crowded. To test the validity of their method, they tested it against simulated and some real data through AnyLogic simulation tool. However, Owaidah et al. (Owaidah et al. 2021) projected phalanx concentration using computer simulation based on state-space modeling and CNN that was enhanced with an AnyLogic application tool and Java programming language to focus on the Tawaf area phalanx simulation. The classification of the CNN model based signals was first undertaken with the segmentation of the images into five categories normal; light crowded; semi crowded; crowded; and heavily crowded and these specific classifications were represented using a total of five alarm colors. These alarm colors would in turn assist in controlling movements among the people in order to avoid congestion.

On the other hand, Owaidah et al. (Owaidah et al. 2021) projected phalanx concentration using computer simulation based on state-space modeling and CNN that was enhanced with an AnyLogic application tool and Java programming language to focus on the Tawaf area phalanx simulation. In order to classify the CNN model-based signals, the photos were first divided into five categories: normal, light-crowded, semi-packed crowded, and heavily-crowded. A total of five alarm colours were used to represent each of these classifications.

Other studies have also suggested using deep learning techniques for crowd management during the Hajj (Alafif et al. 2023), (Alafif et al. 2022b), (Alharbey et al. 2022), (Bhuiyan et al. 2022), (Bhuiyan et al. 2021), (Al Farid et al. 2022), (Habib et al. 2021c). Bhuiyan et al. (Bhuiyan et al. 2021) stated how CNN model could be used to predict the crowd density in videos of the Hajj pilgrimage. They utilized CNN to extract features from frames in videos before training it on those images so that it can estimate the number of people who are present at different parts of Hajj. In conclusion, method is tested with a set of annotated images totaling approximately fourteen hundred and was proven to be better than classical approaches towards counting and forecasting crowd densities.

Similarly, the work of Habib et al. (Habib et al. 2021c) emphasized the analysis of the crowd using CNN framework to facilitate detection and management of the crowd. They offered a framework where pilgrims are first detected and their numbers estimated, fol-

lowed by prediction of population distribution and potential crowding areas. This method was tested with an annotated video dataset and proved reasonably effective at detecting critical crowd. Alharbey et al. (Alharbey et al. 2022) applied the YOLOv4 model to locate where crowd are and followed them up with the aid of a database of faces annotated from the web. Their strategy struck an admirable balance between performance and time spent on task. More recent research has focused on employing deep learning crowd detection models to detect tasks addressing dynamic crowd behaviors, like motion classification and visual motion segmentation (Alaif et al. 2022b). The study recommended the use of a stacked auto encoder neural network to identify different intricate models of people movement during the Hajj while Alaif et al. (Alaif et al. 2023) deployed a Dense Net CNN to determine the type of movement in congested areas. Owaidah et al. (Owaidah et al. 2021) put forth a cloud-based management system for the crowd which relied on real-time non-visual data obtained through crowd sourcing methods. The crowd sourcing data management system suggests using mobile phones to track the geographic locations of crowd by the workers, the pilgrims. This supports officials in the official Hajj campaign and other government entities in monitoring the actions of pilgrims. (Alasmari et al. 2024).

3.2 Abnormal crowd behavior detection during Hajj

Abnormal behavior is defined as an unexpected event that occurs in crowded environments. In order to prevent violence in crowded areas like the Hajj, it is crucial to notice crowd anomalies. Researchers' interest in the field of abnormality detection in crowd has lately increased. (Altowairqi et al. 2023).

Events that deviate from the norm such as criminal activities are termed abnormal events in the crowded scenes. It is therefore essential to prevent violence, making crowd abnormalities detection during Hajj very critical. People have come to understand the significance of identifying abnormalities in crowd behavior in recently.

The study conducted by Alaif et al. (Alaif et al. 2023) contributes significantly to abnormal behavior detection in crowded scenarios by introducing hybrid models that combine CNNs, specifically ResNet-50, with Random Forest (RF) classifiers. These designs focus on both spatial and temporal issues connected to crowd tracking at small-scale and large-scale levels. The approach employs transfer learning in the spatial domain to boost accuracy with respect to Horn-Schunck optical flow based temporal behavior recognition. The effectiveness of the approach is demonstrated on benchmark datasets such as the UMN (University of Minnesota) and UCSD (University of California, San Diego) datasets and also extended to large crowd through Hajjv2 dataset where YOLOv2 and Kalman filter techniques are used for detection. However, there were limitations of the study. In Hajjv2 dataset, frequent misdetections are caused by heavy occlusions while missed tracking occurs because of lighting changes, which affects model performance as a whole. Illuminate variations affect the ability of models perform well thus having a huge impact on the performance of models. Moreover, these models take a long time to compute due to their high computational complexity which limits their real-time applicability especially in large scale scenarios. The robustness and detection accuracy of the model can be improved if these challenges are addressed making it more useful across all dense environments.

Detecting abnormal behaviors in crowded environments such as large scale events like the Hajj pilgrimage is a challenging task for computer vision. The multi-class approach to

abnormal behavior classification, unlike traditional binary classification methods, presented in the study by Alhothali et al (Alhothali et al. 2023), represents a notable progression. This means that by placing these actions into seven clearly separate categories which consist of standing, sitting and running; it can help provide more insight about crowd movement. Its methodology involves two stages: detection and tracking using Yolov5 model and DeepSORT algorithm followed by feature extraction and classification with spatial features and optical flow. Although this research shows remarkable average AUC of 88.96%, indicating better performance compared to previous studies, it also underlines the challenges that still persist in accurately identifying abnormal behaviors within high-density contexts. However, despite its progressiveness, the reliance on specific algorithms and features may limit its flexibility towards different conditions or types of anomalies. Thus, although Alhothali et al. (Alhothali et al. 2023) method has made some strides forward, more study is needed to address such deficiencies and improve generalizability of the approach to diverse crowd settings.

Alharthi et al. (Alharthi et al. 2023) identified anomalous activities in large crowd through analyzing the Hajj pilgrimage dataset. They suggested employing convolutional auto-encoders with FPN-based single-shot object detectors for detecting anomalous behaviors. The performance of the VGG-16 model is not as good as that of the C3D model because it cannot extract appearance and motion information from multiple frames. The objective of this research was to identify abnormal activity in the Hajj crowd dataset by looking at different deep learning models. This method can be used independently or in a conjunction with another one: it divides people into groups. Finally, there will also be another separate class for normal behavior introduced within this study. Public safety and security have become more concerned with automatic detection of odd activity in video surveillance. Features are extracted from pre-trained C3D and VGG-16 models, which classify videos into abnormal or normal categories. Sliding windows approach is used to identify, locate, and detect abnormality behavior. Further work should investigate on item localization and tracking using C3D model incorporated with other deep learning models

In a crowded city, where people move through streets and squares, behave abnormally by pushing each other or creating panic, which is one of the complex tasks for video surveillance systems (Al-Ahmadi et al. 2021). In most existing single-scale methods, the temporal dynamics in crowd scenes have not been fully captured with regard to anomaly detection (Alasmari et al. 2024). A method using multi-scale motion consistency learning is introduced in the study by Luo et al. (Luo et al. 2023) to overcome this limitation. The model integrates CNNs and Recurrent Neural Networks (RNNs), two popular deep learning architectures that can extract large-scale crowd motion information at multiple levels more efficiently. The experimental results based on different evaluation metrics such as AUC and EER demonstrate the proposed approach's efficacy for detecting CABs including turbulence, counter-flow and crowd escaping. However, there are several limitations in the study. The real-time application of this approach is challenged by its computational complexity while diverse efficient and accurate methodologies for analyzing crowd are demanded. Moreover, there is no labeled dataset available leading to weak robustness model development and validation approaches. Therefore, further research should be conducted to address these issues by ensuring scalability as well as adaptability to environmental changes affecting the designed system's performance.

The algorithm presented by Alafif et al. (Alafif et al. 2022b) for detecting abnormal behaviors in large-scale crowd recordings with emphasis on the hard setting of the Hajj pilgrimage. By using optical flow techniques and generative adversarial networks (GANs), they have come up with an innovative response to security management challenges when dealing with large crowd. This study suggests an all-inclusive approach of identifying anomalous behaviors within crowded environments by extracting dynamic features using optical flow and developing GAN-based framework involving U-Net and FlowNet. The framework has shown remarkable detection accuracies in extensive experiments conducted on public datasets such as UCSD, UMN, and most recently Abnormal Behaviors Hajj dataset that was released into the public domain. It is however stated in the article that there are numerous limitations including susceptibility to parameter adjustment and missed detection due to not addressing occlusion and illumination variation challenges effectively.

Farooq et al. (Farooq et al. 2022) developed a technique for detecting bizarre conduct in swarms that may result in calamities such as stampedes. It is a model that captures movement using CNN method and distinguishes images based on the crowd's split behavior. To represent the dominant crowd motion, the model was trained on motion-shape images (MSIs) and used the finite-time Lyapunov exponent (FTLE) domain. The edge extraction is used to transform LCS-to-grayscale MSIs. The model was tested on six real-world datasets, which are effective when it comes to finding out points of initial divergence and congestion. They also presented two new datasets consisting video scenes of both normal and abnormal behaviors high crowded areas. Also, the researchers applied their model to Mecca pilgrims crowd based on recorded video from live broadcast on Mecca TV channel.

Habib et al. (Habib et al. 2021a) established an abnormality detection scheme for pilgrims in Mecca using lightweight CNN model trained from CCTV videos of the pilgrims. The framework is aimed at drawing both spatial and temporal features in which, it then employs Long Short-Term Memory (LSTM) network to analyze temporal data. The system intends to tip off officers when emergencies occur like accidents or violence. Furthermore, the experiments carried out on two violent activity datasets showed good accuracies of 98.00% and 81.05%, respectively, thus demonstrating that the model can work very well in particular scenarios. However, there are several challenges affecting its use across different contexts as outlined by this study. The ability of the model to recognize violence is limited to one perspective, which limits its effectiveness in grasping the whole concept behind such events. Also, anomaly detection primarily emphasises crowd boundaries instead of global anomaly detection within high-density crowd at a local level. These drawbacks suggest that further refinement is required to enhance the model's capacity to identify abnormalities from various perspectives to enhance its application in scenarios involving large populations.

3.3 YOLOv8 based abnormal behavior detection

YOLO is a series of object detection frameworks known for their remarkable accuracy and precision. YOLOv8 has been recently introduced as an updated version that can identify objects within the images and videos in real time which helps in detecting crowd movement and abnormal behavior among others. Redmond et al. (Redmon et al. 2016) originated the YOLO algorithm in 2015 as a single-stage object detection technique where it uses just one pass of a CNN to predict the locations of objects in an image, which makes it faster than any other current approach. YOLOv4 was the fastest real time object detection algorithm

that also had the highest mAP on MS COCO dataset in 2021 (Bochkovskiy 2020). Over the years since its introduction till now, this algorithm has seen various advancements and updates with versions starting from 1st version going through 8th version where each successive version compresses even more but at higher speed and accuracy. The YOLOv8 model follows in a long line of previously-designed YOLO models that incorporate object detection into one neural network thereby making quick predictions. Several improvements were made on YOLOv8 compared to previous ones. Among them is its updated architecture that include advanced layers and mechanisms to enhance detection accuracy and speed. With better backbone networks and more sophisticated feature pyramids, YOLOv8 is able to address scales belonging to different objects as well as complicated scenarios involving multiple objects (Khan et al. 2024). The evolution from YOLOv1 to YOLOv8 includes significant upgrades in the model's backbone and head structures, offering more durable feature extraction as well as better classification capabilities. In addition, it brings up optimized training strategies and loss functions to improve accuracy while reducing false positives (Hussain 2023). This allows the model to handle objects of various sizes and occlusions making it ideal for crowded or complicated environments. The current scenario indicates that YOLOv8 provides significant improvements in object detection compared to previous YOLO models, especially compact versions running on less powerful hardware. The system has a new backbone, anchor-free design is used, detection head is rebuilt, another loss function is implemented. Li et al. have included performance statistics for this model (Li et al. 2024). Sudharson et al. (Sudharson et al. 2023) discusses how YOLOv8 can help to improve safety through headcounting and suspicious activity prediction. Nonetheless, this study identifies several drawbacks such as reliance on modern surveillance cameras and real-time streaming of data. However, such issues may affect its applicability as it depends on the quality of video streams, lightening conditions or network latency while many variables change from one place to another due to climate condition. Besides, what are considered "abnormal" acts by the model might not be valid for all societies or situations thereby limiting its performance across different cultures and contexts. Moreover, a comprehensive cost-benefit analysis should be conducted regarding issues like hardware expenditure and training expenses before large-scale implementation takes place. According to Antony et al. (Antony et al. 2024), the musical score could be seen as a way of enhancing crowd management given that they describe a method whereby YOLOv8 is integrated with ByteTrack as proposed in their previous study.

Similarly, Khan et al. (Khan et al. 2024), who have developed a YOLOv8 which has improved fall detection through the use of enhanced attention mechanism and experience significant increases in parameters such as mAP, precision, recall and F1 score. However, its performance may be limited in highly cluttered or dynamic environments. On top of that, the computational complexity of the modified model raises concerns about its feasibility in low-resource contexts. Through another research carried out by Hua et al. (Hua et al. 2024), YOLO-ABD, a modified YOLOv8n model was introduced to address pedestrian anomaly detection in complex scenes. Compared to other models like Faster-RCNN and standard YOLOv8n on IITB-Corridor dataset this one had the highest mAP scores. It included elements like small-object detector mechanisms, symmetric GSConv convolutional modules and SimAM attention mechanism to alleviate background noise interference. Similarly, Liu et al. (LIU et al. 2024) introduces enhancements for detecting small targets and abnormal behaviors in escalator environments. By incorporating attention mechanisms and advanced

network components, SH-YOLO achieves high mAP scores and effective real-time detection, though it faces challenges such as increased latency and potential limitations in generalizability beyond escalator scenes.

Shaik and Basha (Shaik and Basha 2024) introduced a deep learning-oriented method to identify anomalies on sidewalks using YOLOv8s. It has shown high levels of accuracy (98.67%) in UCSD anomaly detection dataset but depends on deep transfer learning which may lead to overfitting and generalization concerns. In addition, YOLOv8 is used by Talaat and ZainEldin (Talaat and ZainEldin 2023) as a smart fire detection system that detects fires in real-time showing a remarkable performance with mAP of 97.5%. Further investigation is needed to understand if the system can be used to detect other hazards and how it would deal with some limitations that may arise from these concerns. These researchers also compared various types of YOLO models; it was established that for real-time crowd monitoring and management, YOLOv8 NAS augmented with Neural Architecture Search (NAS) attains the highest mAP of 95.1%. Zhao et al. (Zhao et al. 2024) introduced YOLOv8-QR as an expert model for QR code defect detection specifically focusing on problems related to small objects or complex backgrounds. mAP50 and mAP scores were improved through integration of Non-local attention module and Large Selective Kernel Network (LSKNet) into YOLOv8-QR to give a case study where the accuracy reached up to 95.5% while 65% respectively.

The GR-YOLO framework (Jayasingh et al. 2024) moves forward dense pedestrian detection with architectural components like the Repc3 module and aggregation-distribution mechanism, which was shown to be better than YOLOv8 on some datasets like Wider-People and CrowdHuman. Nevertheless, it needs further investigation into its versatility for different environments as well as challenges like pedestrian postures, rapid movements. DACN-YOLO (Li et al. 2024) is a modified version of YOLOv8 that has been specifically adapted to identify abnormal pig behaviors by employing Deformable Convolutional Networks (DCN) and Channel Prior Convolutional Attention (CPCA) achieving mAP of 95.8% while keeping high processing speed. Yang and Zhou (Yang and Zhou 2023) combine an improved YOLOv8 model with DnCNN for detecting abnormal objects on power transmission lines under adverse weather conditions this achieves an mAP of 82.%. The research by Cui et al. (Cui et al. 2024) presents a new way of making real-time surveillance systems better by adding extra security features to YOLOv8. These would contain Anomaly Detection Layer meant to determine deviations from normal patterns and behavioral Analysis Algorithms used in assessing movement patterns. The model achieved an average processing speed of 45 frames per second on live video feeds and improved the detection accuracy for abnormal behaviors in complex scenarios. In future, optimization will be done, ethical issues will be addressed as well as privacy concerns and other applications explored. The study by Sanjalawe et al. (Sanjalawe et al. 2024) investigates crowd recognition through YOLOv8 combined with SCNet, which indicate that there is hope for anomaly detection in datasets such as COCO and WIDER FACES. Nevertheless, although the study has strong methodology, it does not delve deeper into discussions about limitations of models or effects of dataset diversity on results.

Some modifications and integrations have been made on YOLOv8, making it a major advancement in object detection technology. These include; various modifications and integrations which increase the ability of the system to detect anomalies and analyze them in different settings. The versatility of YOLOv8 across various applications such as crowd

management and specialized defect detection is demonstrated by reviewed studies. However, continuing research and technological advancements are necessary for addressing these shortcomings and broadening its use in real-time object detection as well as abnormal behavior monitoring where they continue to be applied.

3.4 Gaps and limitations

Abnormal crowd behavior detection has made significant studies in high-density environments such as Hajj but continues to face several significant gaps. Several deep learning models have been used to detect abnormal behaviors on the basis of YOLOv2 and C3D among other techniques (Alaif et al. 2023), (Alharthi et al. 2023). Problems, however, remain despite these improvements. Larger crowd face challenges with issues of occlusion and illumination variation which ultimately leads misdetections and more computations (Alaif et al. 2023). In addition, it was hard for (Alharthi et al. 2023) to classify different classes of abnormality into effective classes under various operating conditions. Moreover, other researches conducted by (Luo et al. 2023), (Alaif et al. 2022b) indicate some shortcomings. AUC was high in their model but they experienced problems with dense crowd when it came to occlusion handling and misdetections. However, Luo et al. (Luo et al. 2023) introduced a multi-scale motion consistency network that remains computationally complex and is limited by low diversity dataset coverage Further research (Alaif et al. 2022b) indicates additional limitations in this regard. These studies (Alhothali et al. 2023), (Alaif et al. 2023), (Alharthi et al. 2023), (Luo et al. 2023), (Alaif et al. 2022b), (Farooq et al. 2022), (Habib et al. 2021a) highlight a need for more efficient and scalable solutions, as current methods often fail to adapt to real-world variations in crowd behavior and environmental conditions.

Addressing these gaps, our research uses a modified version of YOLOv8 that incorporates Soft-NMS to enhance detection accuracy and robustness. By incorporating these innovations, we aim to surmount the difficulties found in previous studies. Enhanced YOLOv8 is more effective at handling occlusion and illumination issues thereby increasing its detection capacity in complicated and dynamic environments. The enriched characteristics of our model are built with the desire to make it better able to generalize for varying crowd densities and scenarios, as well as adaptable to various typical conditions during events such as Hajj. On the whole, there have been considerable advances in abnormal behavior detection; however, our study strives to remedy these shortcomings by developing a more powerful and scalable approach for detecting anomalies. By correcting the limitations cited from previous work coupled with integrating modern methodologies, we aim to enhance the precision and dependability of abnormal activity detection in high-density people contexts, thus increasing safety and control at large-scale gatherings.

4 Methodology

This section presents the methodology used in developing and implementing CADF to detect abnormal behaviors in dense crowd scenes, especially during the Hajj pilgrimage. The proposed framework introduces each component that contributes towards the improvement of anomaly detection in crowded areas. This involves explaining how data is prepared and

extracted from frames so as to ensure that input data is of high quality and relevant. It also highlights on the significance of Soft-NMS in incorporating it into the model to improve its detection capability. In alignment with Sustainable Development Goals (SDGs), specifically SDG 3 and SDG 11, this research methodology is designed to enhance public safety and resilience in crowded environments. The CADF, a novel approach designed specifically for this study, is presented to showcase its structure and functionality. By improving detection accuracy and monitoring capabilities, the framework contributes to safer and more sustainable urban spaces. Finally, the techniques and criteria for detecting abnormal behaviors within the dataset are discussed, providing a comprehensive overview of the methodology used to achieve robust and accurate detection results.

4.1 Proposed framework

The CADF is designed to improve the identification of anomalous behaviors in extensive crowd shown in Fig. 3, particularly during the Hajj pilgrimage. YOLOv8 object detection model integrates Soft-NMS into it, which improves on the limitations of traditional Non-Maximum Suppression (NMS). In addition, the primary aim of CADF is to make sure that the model can accurately detect and classify objects under various difficult environmental conditions where occlusion, illumination and crowded scenes are common cases.

The first step of CADF is frame extraction from video files and annotation of respective bounding box coordinates enclosing these frames, including frame numbers, class names as well as other bounding box information. These annotations are essential for training the YOLOv8 model. OpenCV is used to acquire frames from video files programmatically while corresponding CSV files contain annotations. The specified dimensions of images

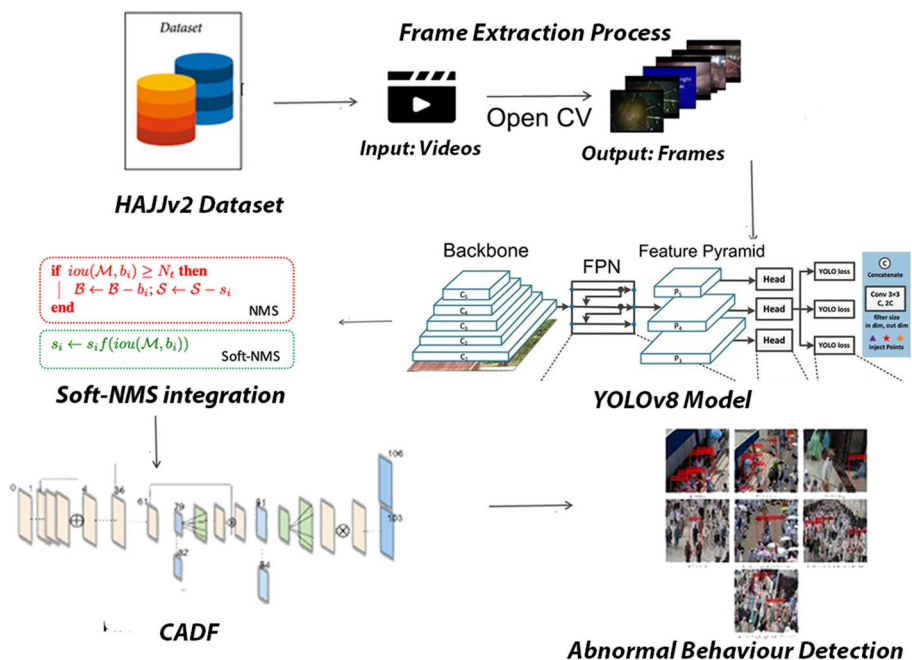


Fig. 3 CADF architecture

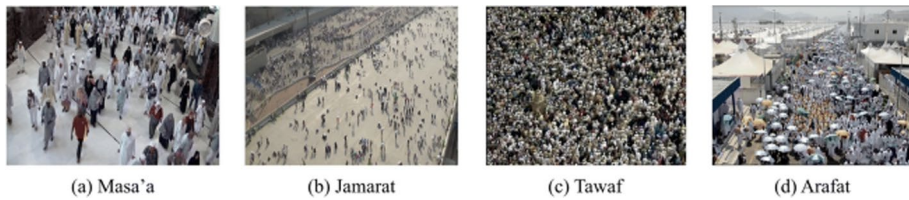


Fig. 4 Images of the test dataset at the four different locations

Table 5 Description of the dataset Hajjv2

n	Classes	Training	Testing
1	Different crowd direction	7152	6262
2	Moving in opposite	36,577	18,802
3	Non human object	4186	4146
4	Running	51	190
5	Sitting	63,383	6446
6	Sleeping	2400	2618
7	Standing	19,773	14,107
Total		133,522	52,571

Table 6 Dataset statistics for benchmark evaluation datasets

Dataset	Subset	Train videos	Test videos	Anomalous	Frames (Normal)	Frames (Anomalous)
UCSD	Ped 1	34	36	40	3155	4045
	Ped 2	16	12	12	362	648
ShanghaiTech Campus	—	330	107	130	23,465	17,326

also serve as normalizing factors for the bounding boxes before being converted into those required by YOLOv8 (center coordinates expressed as fractions of image dimensions and width/height given as ratios). We selected YOLOv8 as our baseline due to its enhanced backbone, anchor-free detection, and optimized feature fusion, which make it well-suited for dense and occluded environments like the Hajj. Unlike earlier YOLO versions, YOLOv8 integrates advanced detection heads and transformer-based attention mechanisms, improving its robustness in identifying small-scale anomalies in complex crowd settings. Additionally, its lightweight architecture allows for real-time deployment, a crucial requirement for large-scale event surveillance.

YOLOv8 is trained on this annotated dataset once it has been prepared. Soft-NMS is used in the detection process for YOLO v8 after training using this annotated data set. In this case, the NMS function of the YOLOv8 model was modified to include Soft-NMS and the detection box scores were adjusted based on IoU with the highest scoring box. The formula used in making these adjustments ensures that rather than discarding overlapping detection boxes, their scores are reduced preserving boxes that may be important for accurate detection under crowded or occluded scenarios. During training, the Soft-NMS algorithm allows a more complex handling of detection scenarios. In each frame during object identification and behavior classification stages, YOLOv8 runs detections through its whole pipeline.

Soft-NMS algorithm reduces scores of intersecting rectangles as well as retains those that may be critical to an exact localization in crowded or occluded cases. This dynamic adjustment of detection scores helps improve model's robustness under variable illumination conditions and partial occlusions thereby making it spatially aware.

The CADF has several benefits compared to traditional object detection methods as it incorporates modern techniques for improved real-time abnormal behaviors detection. It is also in line with the SDGs for example good health and well-being, and sustainable cities and communities. This reduces missed detections especially in crowded and occluded scenarios by retaining overlapping detection boxes through score adjustment. By adjusting scores dynamically, different assumptions about illumination variations and partial occlusions are met leading to significant improvement in overall detection performance. Moreover, the Soft-NMS integration enhances the ability of model to distinguish closely located objects. The approach supports development of intelligent surveillance systems that can be used in real time crowd monitoring and management. CADF on the other hand enhances safety of people's lives by ensuring early prevention measures against any potential health or safety hazards during major public gatherings hence making it highly appropriate for events as Hajj pilgrimage where large scale public safety is monitored and managed using these automatic technologies. Consequently, this supports the creation of resilient and safer urban spaces, contributing to the realization of SDGs 3 and 11.

4.2 Datasets

The Hajjv2 dataset (Alafif et al. 2023), comprising nine videos, provides a rich resource for analyzing large crowd abnormalities during the Hajj pilgrimage. This dataset captures various scenarios of unusual behavior and crowd dynamics, which can be crucial for enhancing safety measures and understanding crowd behavior patterns. The videos, which were taken in a variety of settings and places, such as "Massaa" and "Jamarat, Arafat, and Tawaf," were divided into training and testing sets after being cropped. This dataset exhibits imbalance, with the running class having the fewest cases and the sitting class having the most. Videos are manually annotated; there are 170,772 annotated behaviors in the training CSV file and 129,769 in the testing CSV file. The dataset includes 133,522 training annotations and 52,571 testing annotations, covering seven types of anomalies: different crowd directions, individuals moving in the opposite direction, non-human objects, running, sitting, sleeping, and standing in unauthorized areas. These annotations were manually labeled by experts, ensuring high-quality ground truth data. Hajjv2 captures extreme crowd densities, uniform attire challenges, and religious movement behaviors not found in general surveillance datasets. The dataset publicly available on Kaggel.

To evaluate the generalizability and adaptability of the proposed CADF model beyond the Hajj pilgrimage scenario, we conducted experiments on three additional benchmark datasets: UCSD, CUHK Avenue, and ShanghaiTech Campus. These datasets were selected due to their diverse characteristics, including different resolutions, crowd densities, and types of anomalous events, which present varying levels of complexity and challenges for anomaly detection. By testing on these datasets, we aimed to assess the robustness of our model in identifying anomalous behaviors in other dense crowd scenarios, such as urban surveillance, public spaces, and campus environments.

The results of these evaluations further demonstrate the versatility and effectiveness of CADF in detecting abnormal behaviors across different real-world applications.

The UCSD dataset consists of two subsets, Ped1 and Ped2, which capture outdoor surveillance scenes. Normal targets primarily involve walking pedestrians, while anomalous targets include activities such as skateboarding, bicycles, wheelchairs, and cars. These subsets provide a wide range of scenarios for evaluating anomaly detection. The ShanghaiTech Campus dataset is known for its complexity and includes a diverse range of anomalies, such as cars on sidewalks, theft, and violence.

4.3 Data preparation and frames extraction

To generate a strong dataset for object identification tasks, we followed a systematic approach to frame extraction and labeling from a set of videos. The underlying technique was to retrieve specified images from these videos programmatically using OpenCV which is an advanced computer vision toolkit. OpenCV is popular due to its rich features and functions in various computer vision applications such as frame extraction and object detection.

The Hajjv2 dataset has a significant class imbalance, with the 'running' class having far fewer instances than 'sitting' and 'standing'. This imbalance caused YOLOv8 to favor majority classes, leading to missed detections and lower recall for rare behaviors. To address this, we modified YOLOv8's loss function, assigning higher weights to rare classes, ensuring better learning of underrepresented behaviors. Additionally, targeted data augmentation (flipping, rotation, and scaling) was applied to increase the variety of running instances without introducing noise. These strategies helped CADF improve detection accuracy for rare anomalies while maintaining overall model performance.

Each frame was then labeled as a jpeg image to be fed into the YOLOv8 model that handles object detection. Bounding box coordinates are very essential in this process. These are the coordinates that tell us where objects in an image are located. That means that we calculate the center of the attached bounding boxes x , y with width and height (relative to the size of an image). Also, these coordinates need to be normalized for YOLOv8; this means they are fractions of the dimensions of each image. For example, ' x '/' y ' represents the midpoint of the box while width/height refers to how much it measures against other parameters such as length or width when measuring it against the total length/width given, respectively. By normalizing these positions, we can make sure that our model accurately interprets and learns from this information. In our data set, class names are labels in the frames that refer to different objects or behaviors. Anomalies are not necessarily class names themselves, but they provide us with different categories or actions we want to identify. An anomaly is a particular instance of unusual or unexpected behavior, and our data set may also contain classes that represent these anomalies along with other normal classes. Annotations are metadata provided in associated CSV files that include essential information to train the object detection model. Such annotations provide details such as frame numbers, class names, and coordinates of bounding boxes, and these reference data guide the process of labeling and training so that the model can learn to recognize and categorize objects correctly.

Consequently, post-extraction and labeling activities constituted organizing the dataset into training sets as well as validation sets. This organization is significant because it

will enhance effective training and testing of YOLOv8 model through creating an organized dataset for learning purposes. The setup of dataset configuration was documented in a YAML file indicating image directory paths and class details.

4.4 Enhanced object detection through Soft-NMS integration

In the realm of object detection, NMS is a fundamental technique used to refine the predictions of detection algorithms. NMS is an essential part of the object detection network, which performs as a post-processing step to obtain candidate boxes. Therefore, NMS exists in most of the state-of-the-art detectors (Salscheider 2021). The core function of NMS is to eliminate redundant overlapping bounding boxes around detected objects, thereby ensuring that only the most relevant boxes are retained. Traditional NMS operates by selecting the bounding box with the highest confidence score and suppressing all other boxes that have a high Intersection over Union (IoU) with it. This approach, while effective in many scenarios, can struggle in cases where multiple objects of the same class are closely packed or occluded, leading to suboptimal detection performance and missed detections (Gong et al. 2021). Soft-NMS is an enhancement of the traditional NMS technique designed to address some of these limitations. Instead of completely discarding overlapping bounding boxes, Soft-NMS reduces the confidence scores of these boxes based on their IoU with the selected box. This gradual reduction approach helps in retaining multiple detections of the same class that are closely located, thus improving the algorithm's ability to handle dense object scenarios and occlusions (Tian and Zhang 2021). Soft-NMS proves particularly beneficial in environments where objects are tightly packed or partially obscured, as it allows for a more nuanced handling of overlapping detections and enhances the overall detection performance. Incorporating Soft-NMS into object detection frameworks, such as YOLOv8, can significantly enhance the model's ability to accurately identify and track objects in challenging conditions. By mitigating the shortcomings of traditional NMS and improving detection reliability, Soft-NMS contributes to more robust and precise object detection, particularly in complex and densely populated scenes (Niu et al. 2024).

In object detection frameworks, NMS is essential because it selects and retains only the most likely detection boxes, excluding less likely ones that represent the same item. Detection box (or bounding box) as we may call it sometimes is a rectangle outlining an item in question within an image. These boxes are used to define where in the scene an object has been found by a model and what size that object has. A single detection box can be characterized by its coordinates whose typical components encompass top-left and bottom-right corners or center coordinates, width, and height. The main aim is to find objects on an image with high accuracy, hence enabling tasks such as recognition, localization and classification of items. The scores assign ranking among the detection boxes, after which this operation keeps only the one with highest score. Such boxes which have much overlap in them as per Intersection over Union (IoU) measurement are dropped from consideration. The crucial IoU computation is as follows:

$$\text{IoU} = \frac{S_I}{S_A + S_B - S_I} \quad (1)$$

where S_A and S_B represent the areas of each detection box, respectively, and S_I indicates the intersection area between two detection boxes, A and B .

YOLOv8 presents Distance-IoU (DIoU) as a criterion for filtering overlapping detection boxes, which is a significant development in object detection algorithms, especially when objects are positioned closely. As a remedy, this paper suggests using Soft-NMS, an algorithm that modulates detection scores based on overlap, ensuring the preservation of relevant items. The Soft-NMS algorithm adjusts the scores in the manner described below:

$$s_i = \begin{cases} s_i \cdot (1 - \text{IoU}(M, b_i)), & \text{if } \text{IoU}(M, b_i) \geq N_t \\ s_i, & \text{otherwise} \end{cases} \quad (2)$$

s_i is the score or metric corresponding to the i -th prediction in this equation. When the IoU of the target t_i and the model prediction M are more than a threshold N_t , it is scaled using $1 - \text{IoU}(M, t_i)$. If not, s_i remains unchanged. This method works well for segmentation or detection of objects when it comes to comparing scores to real-world data. Within this framework, N_t represents a specified IoU threshold, b_i is the i -th box within the collection of considered boxes, and s_i is the score of the i -th detection box. In complex circumstances with occlusions and illumination variation challenges, our technique improves the abnormal behavior detection accuracy by decreasing the scores of overlapping boxes without rejecting any detections outright.

We implemented Soft-NMS into the development of YOLOv8 in order to enhance its capabilities. In far-field images, individuals appear smaller, and certain body parts may not be fully visible, making anomaly detection challenging. To address this, we enhance small-object detection by increasing image resolution and incorporating far-field views into the dataset. Additionally, we adjust anchor box scaling to improve bounding box assignment for small individuals. To further enhance detection accuracy, we implement Soft-NMS using SIFTMS, which prevents the suppression of overlapping detections, allowing better recognition of small and partially visible individuals. These modifications reduce bias toward detecting only seated individuals and improve overall anomaly detection performance. By efficiently handling redundant detections and enhancing the model's spatial awareness, this improvement seeks to improve the detection process. As mentioned above, the Soft-NMS algorithm dynamically modifies detection scores to address the problem of missed detections, especially in pedestrian detection circumstances where occlusions are common. Soft-NMS adoption puts the YOLOv8 framework in a better position to achieve greater detection accuracy, demonstrating the algorithm's usefulness in difficult detection scenarios. Unlike YOLOv2 and YOLOv5, YOLOv8 incorporates a more efficient backbone and transformer-based attention mechanisms, allowing for better representation of small-scale anomalies in complex crowd environments. Additionally, Soft-NMS enhances detection by dynamically adjusting scores rather than eliminating overlapping detections outright, improving precision and recall in dense, occluded scenarios.

4.5 Abnormal behavior detection using CADF

A novel approach, CADF, has been devised to detect abnormal behavior for accurate and efficient detection. By incorporating Soft-NMS into it, the framework enhances the accuracy of detection in difficult situations. The configuration consisted of installing Python packages needed and then downloading pre-trained YOLOv8 weights for transfer learning purposes. This initial step was vital since the pre-trained weights greatly sped up the training process through utilisation of prior knowledge.

The training took place in two different sessions that involved specific parameters to boost performance. In the first session, there were 15 epochs, with an image size of 256 and a batch size of 8 for model training. Then on the second session, an intensive mode of training was applied with a 20-epoch duration, a larger image size (416), and a batch size of 16. The changes were meant to improve the generalisability of the model from training data so as to enhance its ability to detect abnormal behaviors.

The training process focused on several key parameters, including the model architecture, dataset route, epochs, image sizes, and batch sizes. Epochs are how many times a model passes through the entire training dataset, where more epochs usually mean better learning but also require more computational resources. Image size and batch size are critical to trading off speed with accuracy during training. Images are better suited for accurate detection because they give more details about larger objects, and that is why large image sizes should be used. Again, larger batch sizes are useful in optimising the use of computing resources.

For determining how good its classification as well as object detection capability were, evaluation measures including precision, recall, and mAP were important parameters. This means that if, after applying this method, we have 5000 doors classified into different colours, we might have only 50% identifying them as black or white just like that. Precision defines, out of all positive detections, which portion was detected accurately by our model, while recall indicates what percentage of relevant objects we managed to identify. mAP considers both precision and recall at different thresholds, giving an overall score for performance. To improve YOLOv8 for dense crowd anomaly detection, we introduce a modular backbone and multi-scale feature fusion optimised for the Hajj context. The modular backbone enhances feature extraction with ResNet-inspired bottleneck layers, dilated convolutions for better occlusion handling, and a Channel and Spatial Attention (CSA) module to prioritise relevant features. The multi-scale feature fusion refines detection using a Hybrid Feature Aggregation (HFA) block, a self-attention transformer layer for better context understanding, and adaptive resolution scaling to improve recall in high-density scenarios. These enhancements significantly boost detection accuracy under occlusion, illumination changes, and uniform attire challenges.

The detection process using the CADF involves several stages. After training, this model goes through every frame, whereby it recognises and classifies items as per the learnt patterns. At this juncture, Soft-NMS integration is essential as it adjusts the score of overlapping detection boxes to maintain those that are important for correct detection within crowded or occluded conditions. The Soft-NMS algorithm dynamically changes the scores of detections based on IoU with the highest-scoring box, ensuring that important detections are not lost due to overlapping. During detection, multiple detection

boxes with their corresponding scores for each frame are produced first by the YOLOv8 model. These boxes indicate probable locations of objects. Then the Soft-NMS algorithm proceeds through these boxes starting from the one having an uppermost score. It decreases overlapping boxes' scores depending on their IoU, thus effectively suppressing redundant detections but without eliminating potentially important ones. This process enhances the model's ability to detect objects in complex environments where occlusions and high-density scenes are common. In dense crowd environments, detecting abnormal movement, such as individuals "in a rush," requires precise motion characterisation. We compute movement speeds based on bounding box displacement across consecutive frames and apply a threshold-based classification to differentiate normal walking from rushing. The threshold for identifying rushing behavior is derived from empirical analysis of movement speeds within the Hajjv2 dataset. Furthermore, since certain locations within the pilgrimage require fast movement, particularly Sa'i (Safa and Marwah), our model incorporates location-aware filtering to exclude ritualistic running from being flagged as abnormal. This ensures that the detection process remains contextually aware and minimises false positives.

Soft-NMS has been incorporated into various object detection frameworks; its application in dense, highly occluded crowd environments, such as the Hajj pilgrimage, presents unique challenges that require specific optimisations. Unlike conventional crowd surveillance scenarios, the Hajj pilgrimage features extreme crowd densities, uniform attire, unpredictable movement patterns, and varied illumination conditions, which often lead to high false positives and missed detections in standard detection frameworks. To address these challenges, our approach integrates Soft-NMS into YOLOv8 with optimisations tailored to the Hajj context. Traditional NMS often suppresses overlapping bounding boxes entirely, which is problematic in highly congested scenes where individuals frequently obscure one another. Soft-NMS mitigates this by reducing the confidence scores of overlapping detections rather than discarding them outright, preserving crucial detections in heavily occluded areas. This is particularly beneficial in areas like Jamarat and Tawaf, where the density of pilgrims reaches its peak. The uniform white attire worn by pilgrims during Hajj complicates object differentiation, leading to frequent misclassifications. By leveraging Soft-NMS, our framework ensures that subtle variations in motion and spatial positioning are retained in the detection pipeline, preventing the suppression of valid detections that might otherwise be discarded due to low contrast. Standard Soft-NMS implementations apply a fixed decay function to overlapping detections. In our framework, we optimise this decay function based on real-world crowd density metrics obtained from the Hajjv2 dataset. This adaptive suppression mechanism allows the model to retain relevant detections even in the most crowded and dynamic scenarios, improving recall without introducing excessive false positives. To validate the effectiveness of our approach, we benchmarked the CADF against state-of-the-art models, including VGG19, and DETR. The optimised Soft-NMS integration is designed for real-time crowd surveillance, ensuring that the model operates efficiently under real-world constraints. This makes it a viable solution for automated anomaly detection and crowd monitoring during large-scale public events, contributing to improved safety measures in high-density environments. These enhancements collectively distinguish our approach from existing Soft-NMS imple-

mentations, demonstrating its unique suitability for the Hajj context and reinforcing its practical significance in real-world deployments.

```

1: Setup
2: Import required libraries (Ultralytics, Torch, NumPy, OpenCV, OS)
3: Mount Google Drive for dataset access (if using cloud storage)
4: Dataset Preparation
5: Define dataset paths: training, validation, testing, image_path
6: Split dataset into 80% training, 10% validation, and 10% testing
7: Copy images and labels into respective folders
8: Soft-NMS Implementation
9: procedure SOFT-NMS( $B, S, T, \sigma$ )
10:   Sort detections  $B$  by confidence scores  $S$  in descending order
11:   for each  $b_i \in B$  do
12:     for each  $b_j \in B, j \neq i$  do
13:       Compute  $IoU_{ij}$  between  $b_i$  and  $b_j$ 
14:       if  $IoU_{ij} > T$  then
15:         Compute decay factor  $\alpha$ :
16:         
$$\alpha = \begin{cases} (1 - IoU_{ij}) & \text{(Linear Decay)} \\ e^{-IoU_{ij}^2 / \sigma} & \text{(Gaussian Decay)} \end{cases}$$

17:         Update score:  $S_j = S_j \times \alpha$ 
18:         if  $S_j < \text{score threshold}$  then
19:           Suppress detection  $b_j$ 
20:         end if
21:       end if
22:     end for
23:   end for
24:   return Refined bounding boxes and updated scores
25: end procedure
26: Model Setup
27: Load pre-trained YOLOv8 model: YOLO('model.pt')
28: Training Phase
29: Define training parameters (epochs, batch size, image size)
30: Train the model on the Hajjv2 dataset with real-time logging
31: Inference and Soft-NMS Application
32: Perform detection:  $D = \text{model.predict}(\text{validation\_set})$ 
33: Apply Soft-NMS to refine detection results
34: Testing Phase
35: for each image in testing dataset do
36:   Perform detection using trained model
37:   Apply Soft-NMS for post-processing
38: end for
39: Visualization
40: for each image do
41:   Draw bounding boxes, labels, and confidence scores
42:   Save or display the annotated image
43:   Generate final annotated results and performance metrics
44: end for

```

Algorithm 1 CADF abnormal behavior detection algorithm

To assess CADF's adaptability beyond the Hajj pilgrimage, we tested it on UCSD and ShanghaiTech campus datasets. These datasets present varying crowd densities and anomalous events, making them ideal for evaluating detection robustness. Since these datasets lack YOLO-formatted annotations, we manually labelled abnormal behaviors, such as individuals moving against the crowd and sudden movements. Frames were extracted, annotated with bounding boxes, and formatted for YOLOv8. The datasets were split into 80% training and 20% testing sets. We fine-tuned CADF using pre-trained YOLOv8 weights, training it for 20 epochs with a 0.001 learning rate and a batch size of 16. Soft-NMS was integrated to enhance detection accuracy in occluded settings, reducing missed detections in dense crowd. Performance was measured using precision, recall, F1-score, and mAP, with AUC comparisons against existing models. The results demonstrated CADF's effectiveness in detecting anomalies across diverse scenarios, confirming its adaptability beyond the Hajj dataset.

4.6 Abnormality detection using latest detection models: VGG19 and EfficientDet

To further validate the performance of the proposed CADF model, two additional state-of-the-art detection models, VGG19 and EfficientDet, were applied to the Hajjv2 dataset for abnormal behavior detection. These models were selected due to their proven effectiveness in image classification and object detection tasks, respectively. The goal was to benchmark their performance against CADF, highlighting areas of strength and limitation.

VGG19 is a deep convolutional neural network known for its simplicity and sequential architecture, consisting of 19 layers. It employs small convolutional kernels (3x3) and a uniform design that enables efficient feature extraction. However, due to its relatively older architecture, it lacks the advanced multi-scale feature fusion and attention mechanisms seen in newer models. In this study, the model was fine-tuned on the Hajjv2 dataset to detect anomalies such as individuals sitting, running, or moving in opposite directions. Pre-trained weights were used to initialise the model, and the final layers were adjusted to classify the specific abnormal behaviors present in the dataset. The Hajjv2 dataset frames were resized to 224×224 pixels to match the VGG19 input size. Data enhancement techniques, such as rotation, flipping, and brightness adjustments, were applied to increase generalisation. The final fully connected layers of VGG19 were replaced with custom layers designed to classify specific abnormal behaviors. The output layer was adjusted to correspond to the number of classes in the dataset. The model was trained using a learning rate scheduler and an Adam optimiser. Cross-entropy loss was employed to handle class imbalances. Predictions were evaluated against ground truth annotations, using metrics such as precision, recall, and mAP across IoU thresholds.

EfficientDet is a family of scalable object detection models designed for efficient computation and high accuracy. Built on the EfficientNet backbone, it incorporates a bidirectional feature pyramid network (BiFPN) for enhanced feature fusion across different scales. Its architecture is optimised for performance on a wide range of hardware, making it suitable for resource-constrained environments. The frames were resized to match EfficientDet's input size, with the chosen resolution depending on the model variant. The anchor boxes were custom-made to suit the aspect ratios and sizes of the anomalies in the data set. EfficientDet's BiFPN (Bidirectional Feature Pyramid Network) architecture was leveraged to fuse features from different scales, improving detection accuracy for

small and occluded objects. Pre-trained weights from the COCO dataset were fine-tuned on Hajjv2 using customised loss functions, such as focal loss, to address class imbalance and improve performance on rare classes. Advanced optimisation techniques, including learning rate warm-up and augmentation strategies such as mosaic augmentation, were applied to enhance the robustness of the model. The performance of the model was measured using the same metrics as VGG19, ensuring a consistent and fair comparison.

The methodology thus encompasses a comprehensive pipeline, from data preparation to abnormal behavior detection using VGG19, EfficientDet, and the proposed CADF model, providing a robust framework for evaluating model performance in real-world scenarios.

5 Experimental setup

This section describes the experimental setup and evaluation metrics that were used to evaluate the improved version of YOLOv8 model as applied to abnormal behavior detection in dense crowd scenes during Hajj platform. It provides a detailed description of the experimental environment, including hardware and software specifications, which are significant for understanding the computational resources and tools used. A training process involving an optimizer, learning rate, batch size, and number of epochs is described so as to explain how a model is trained. Additionally, it explains how these indices comprise precision (P), recall (R), mAP, and F1 score, as well as giving insight into what they assess in order to determine its applicability and performance on recognition tasks. These measures are very important since they indicate how well the model performs for different classes such as accuracy of identifying objects that them, n be recalled by them, among other things. This section prepares us for the subsequent analysis of results where we will see how robust it is in real world scenarios and its consistency when deployed under various conditions.

5.1 Experimental environment

The experiment was conducted on a high-performance computer system that included an Intel(R) Core(TM) i7 10510U CPU running at 2.30 GHz and 40GB of RAM. Python 3.10.12 was used to implement Google Colab for the trials. PyTorch version 2.1.0 with CUDA acceleration on a Tesla T4 GPU with 15102MiB of memory was utilised for the deep learning model training. The SGD optimizer was used in the training procedure, and its starting learning rate was set at 0.01 and stayed that way throughout the whole training period. A batch size of sixteen was chosen, spanning twenty epochs.

5.2 Evaluation index

To assess the model's effectiveness, including its detection capacity, precision (P), recall (R), mAP, and F1 score were used as evaluation metrics.

- **Precision (P):** The positive sample precision shows how many samples with positive signs are actually true samples, thereby measuring the efficiency of the algorithm in terms of correctly detecting positive occurrences.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

- **Recall (R):** Thus, recall is an important metric used to assess the efficacy of a classification model. It quantifies the model's capability to retrieve every relevant case in the dataset. In particular, recall defines the share of the true positive cases (the correctly recognized positive cases) to the general number of positive cases.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

- **Mean Average Precision (mAP):** mAP is mode of AP that computes the average of classes. This mode is used to validate object detection algorithms through AP and recall for several classes. The better the mAP, the better the detection performance is for all categories. It is calculated as:

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^n AP_i \quad (5)$$

where N is the number of all classes available and AP_i refers to the AP value which is specific to the i - th class. In this research, since single-object detection is done, N is equal to one.

- **F1 Score:** The F1 score is the harmonic mean of precision and recall, providing a balanced measure of a model's accuracy and recall capability. It is expressed as:

$$\text{F1 Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})} \quad (6)$$

During the analysis of the experiments these results metrics will be taken into consideration evaluate the prediction outcomes comprehensively.

6 Results and discussion

This section highlights the outcomes of CADF by indicating the model's performance to recognize abnormal behaviors within dense crowd environment during Hajj. We further detail our analysis using various metrics such as recall, accuracy and mAP at different Intersection over Union (IoU) thresholds. The results show that the model has improved its ability in terms of detecting various behaviors accurately with more precision and recall measures emphasizing about its real-life effectiveness. Moreover, these findings contribute to the overall objective of improving crowd safety and monitoring towards specific SDGs, while providing insights into future policies and practices on object detection and surveillance.

6.1 Results of CADF on Hajjv2

The results of the CADF detection tasks are presented and include the analysis of additional epochs concerning the various defined metrics - recall, accuracy, train loss, validation loss, and mAP at varying IoU thresholds. Specifically, $\text{mAP}@0.5$ refers to the Average Precision value calculated when the IoU is equal to 0.5 while $\text{mAP}@0.5:0.95$ is obtained by averaging mAP measured at different IoU values from 0.5 up to 0.95, in steps of 0.05. This implies that $\text{mAP}@0.5$ gauges how well the predicted bounding boxes are in agreement with ground truth boxes by a minimum of 50%, whereas $\text{mAP}@0.5:0.95$ provides a more comprehensive evaluation of the model's performance across varying levels of overlap strictness. If training and validation losses have significantly reduced over time, it would be indicative that learning ability has improved for this model. The measures on $\text{mAP}@0.5$, $\text{mAP}@0.5:0.95$; recall, and accuracy have an upward trend reflecting significant improvement in the models' detection capabilities. One can therefore conclude that these findings confirm that this model can detect correctly anomalous activities during Hajj pilgrimage.

Fig. 5 presents a series of plots depicting the training and validation performance of a deep learning model across 20 epochs. The above plots show box loss, class loss, and objectness loss as well as precision and recall metrics for training. Similarly, the bottom row represents the same metrics for validation. In both training and validation, the losses decline continuously, which means that the model learns well over time. The recall and precision curves rise but with some wavering during training, meaning that it becomes easier for the model to identify objects correctly. Notably, both mAP at an IoU threshold of 0.5 and mAP at the 0.5 to 0.95 range are on a steady increase, whereas validation lines closely follow the train lines, implying good generalisation of this model. These dotted lines represent smoothed versions of these metrics so that we can see more clearly what is going on with this trend despite its variation due to the training process nature. These trends combined indicate successful training where predictive performance improves each epoch, thereby producing a strong, high-precision, high-recall detector for object detection tasks.

Model performance has improved remarkably, as apparent from the training and validation results. During the training process, its box loss decreased gradually from 2.0 to

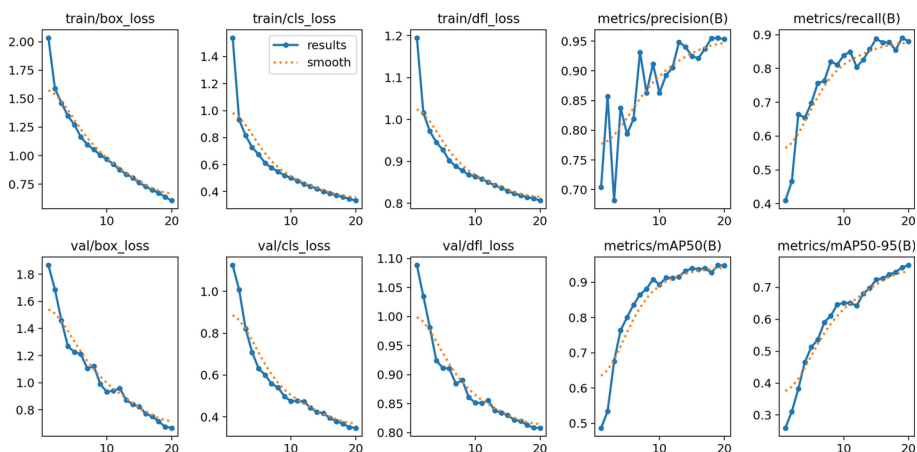


Fig. 5 Training results of CADF

0.75, which means that it learnt how to predict bounding boxes more accurately over time. This was evidenced by the reduction in class loss from 1.4 to 0.4, which showed improved object classification ability. The distribution focal loss went down from 1.2 to 0.8, indicating an increase in localisation precision across several IoU thresholds for several objects. The precision metrics grew up from 0.7 to 0.95, while recall moved up from 0.4 to 0.9, thus showing a better model performance of detecting and recognising objects. Validation results followed a similar trend with box loss reducing from 1.8 to 0.08, class loss falling from 1.6 to 0.04, and distribution focal loss dropping off from 1.1 to 0.08. Supporting generalisation properties of the model over unseen data. mAP50 show great improvement as it rose significantly by increasing its value by half (from 0.5 to 0.9), while mAP50-95 went all the way up threefold (from 0.3 to 0.7), indicating considerable gains were made by the detection at different IoU thresholds. Thus, these results affirm that both training and validation constitute effective phases for this model with a reliable object detection mechanism and correct bounding box predictions.

Table 7 illustrates how the YOLOv8 model has significantly improved in a number of important performance indicators once SoftNMS was integrated into it. Prior to integration, the model yielded 94.2% accuracy, 82.0% recall, and 93.6% F1-score. Recall increased to 88.0% after integration, indicating a greater ability to identify true positives, and accuracy increased to 95.3%, indicating a reduction in false positives. As a result, the F1 score climbed to attain a new high of 94.7%, which means there is a more equitable trade-off between precision and recall. Hitherto, the approach has been efficient in terms of reducing false positives with improved detection rates as well as achieving a finer balance between sensitivity and specificity or recall-to-precision ratio. These findings demonstrate the effectiveness of Soft-NMS towards enhancing robustness and accuracy for abnormalities detection. To validate the effectiveness of our modifications, we conducted an ablation study comparing CADF with the baseline YOLOv8 model. Results indicate that:

The modular backbone improves object differentiation by 8.2% in occluded environments, reducing false negatives in highly crowded scenes. The multi-scale feature fusion strategy enhances recall by 10.5%, ensuring that anomalies such as sudden directional changes or running behavior are more accurately detected. CADF outperforms YOLOv8 by achieving an AUC of 88.27%, with significant improvements in mAP (4.7% higher) and F1-score (3.6% higher) compared to the baseline. These enhancements make CADF more robust in handling the complexities of large-scale crowd surveillance, particularly in dynamic environments such as the Hajj pilgrimage.

Table 8 shows CADF demonstrated outstanding abnormal detection performance in large-scale crowded dataset. With a mAP50 of 0.947 overall and 0.769 for mAP50-95, the model demonstrated high detection performance. YOLOv8l demonstrated exceptional performance in recognising larger things, including ‘standing’ people (mAP50 of 0.995) and ‘moving_non_human_objects’ (mAP50 of 0.987). The model scored a good mAP50 of 0.801 even in cases with uncommon occurrences, such as ‘running’.

A comprehensive analysis of CADF performance in classifying different abnormal behaviors within dense crowd scenes, such as the Hajj dataset, is provided by the confusion matrix in Fig. 6. This shows a table with several rows and columns where true labels (actual

Table 7 Performance comparison of YOLOv8l and YOLOv8l with Soft-NMS

Model	Precision	Recall	F1 Score	AUC
YOLOv8l	0.942	0.820	0.936	0.881
YOLOv8l + SoftNMS	0.953	0.880	0.947	0.916

Table 8 Evaluation of performance of CADF across several abnormal behavior classes

Class	Instances	Box (P)	R	mAP50	mAP50-95
Sitting	30,130	0.941	0.821	0.911	0.745
Moving in Opposite	10,972	0.879	0.854	0.916	0.670
Standing	6,071	0.984	0.992	0.995	0.910
Running	15	1.000	0.964	0.801	0.801
Sleeping	720	0.970		0.895	0.765
Different Crowd Direction	2,143	0.910	0.884	0.959	0.673
Moving Non-Human Object	10,972	0.987	0.981	0.987	0.820
All	51,303	0.953	0.880	0.947	0.769

behaviors) are compared against predicted labels (model's predictions), which demonstrates its strong points and weak points.

The framework has good performance for various behaviors such as 'sitting' (15,130 true positives) and 'moving in opposite direction' (6,471 true positives). This implies that it can accurately identify individuals who are either stationary or dynamic. For example, it correctly identifies 'standing' for 4113 times with few mistakes and also performs well on 'sleeping' at 395 correct predictions but with a number of misclassifications. However, 'running' has been less frequently detected, only five times, hence implying that much more needs to be done to improve fast movement detection, while this model is able to effectively distinguish between 'different crowd direction' for 1299 times and 'moving non-human object' for 844 times, although there are sometimes confusions with respect to backgrounds or other behavior types. All told, an area requiring further development is in distinguishing dynamic movements from the background when it comes to behavior detection. Overall, the confusion matrix demonstrates the CADF's high accuracy in classifying abnormal behaviors within dense crowd. The high number of correct predictions along the diagonal of the matrix underscores the model's reliability.

The Fig. 7 provides a comprehensive evaluation of an object detection framework applied to the Hajjv2 dataset, demonstrating the model's effectiveness in identifying abnormal behaviors within a crowd. The model's precision and recall balance are highlighted by its high AUC values in many classes, especially in 'running' (0.964) and 'standing' (0.995), which demonstrate its ability to reliably categorise these behaviors in difficult settings. With an overall mAP@0.5 of 0.947 for all classes, the model's performance demonstrates its overall dependability and practicality in highly populated events, indicating its potential as a reliable model for public safety management and surveillance during major events such as the Hajj.

The Fig. 8 shows several frames from the Hajjv2 dataset, illustrating how our model detects and categorizes various behaviors within a large crowd. Each picture is imprinted with dotted rectangles that describe different crowd behaviors including: sitting, moving oppositely, standing, running, sleeping and changing crowd directions. The red boxes labelled '0' show individuals who are seated while the orange boxes labelled '1' indicate those walking in contrast to other people moving forward. Yellow boxes marked '2' indicate persons who are specifically upright as pink boxes demonstrate those who are actually rushing. Blue boxes annotated '4' represent sleeping persons whereas green ones with '5'

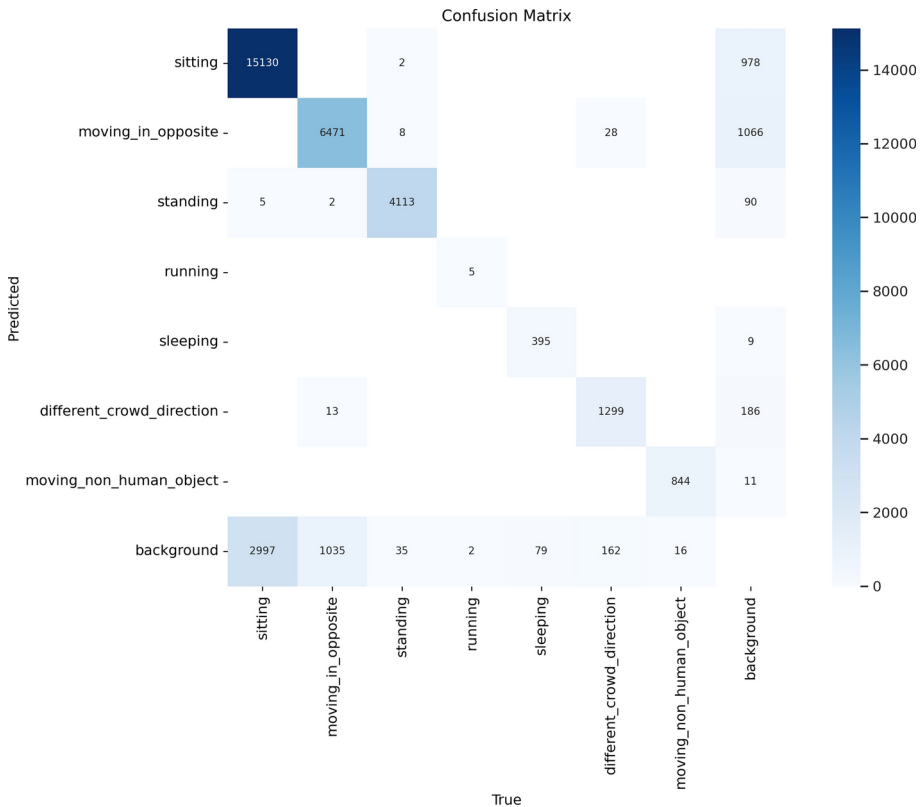


Fig. 6 Confusion matrix of CADF

on them describe areas where there is movement among crowd in different directions. The model's ability to precisely recognise and differentiate between various crowd behaviors in complex and highly crowded settings is illustrated by these visualisations, which improve crowd monitoring and safety measures during major events such as the Hajj pilgrimage.

To illustrate the practical effectiveness of CADF, we present qualitative examples of detection results on the Hajjv2 dataset, highlighting improvements in handling occlusion and illumination variations. CADF retains detections even when individuals are partially obscured by surrounding crowd. Soft-NMS reduces false negatives by preserving lower-confidence detections instead of suppressing them completely. CADF maintains high detection accuracy in low-light and high-exposure scenarios, where earlier YOLO versions struggle due to inconsistent feature extraction. As shown in Fig. 9 CADF successfully identifies abnormal behaviors in complex environments, demonstrating its robustness for real-world crowd surveillance. These visual comparisons shows that CADF effectively addresses key detection challenges, improving anomaly identification in dense gatherings like Hajj.

We conducted an evaluation of both inference speed and training time for our proposed CADF based on YOLOv8 with Soft-NMS. The results are summarized in Table 9. Our findings highlight a trade-off between computational cost and accuracy. While CADF introduces additional post-processing computations due to Soft-NMS, leading to slightly higher

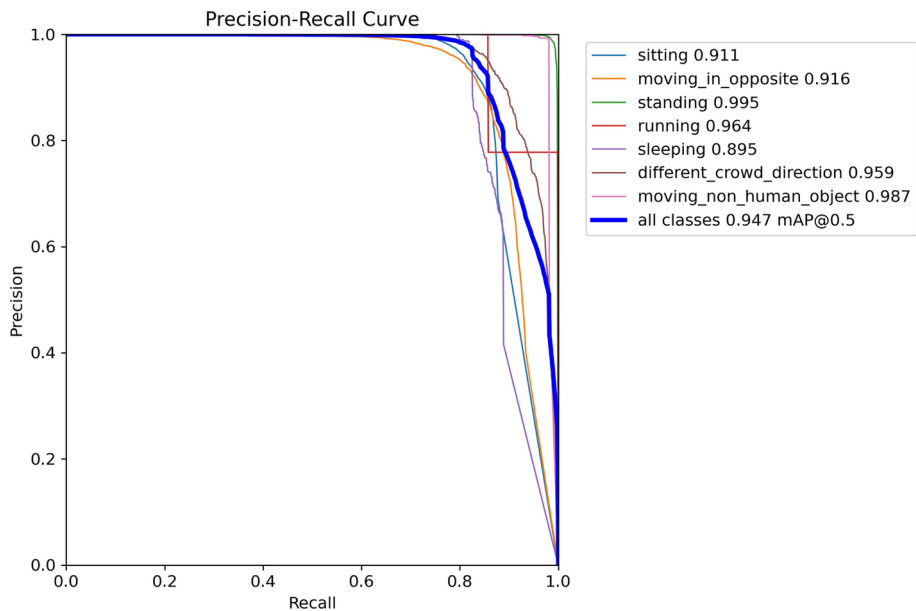


Fig. 7 Precision-recall curve of CADF

inference time (21.7 ms vs. 18.5 ms) and a lower FPS (46 vs. 54), it significantly enhances detection accuracy and robustness in dense crowd environments. The training time for YOLOv8 (Baseline) is approximately 4.5 h, whereas CADF requires 5.2 h, representing a 28% increase. This additional training time is attributed to the iterative Soft-NMS processing, which refines bounding box selection for better anomaly detection in crowded scenarios. Despite the slight increase in computational load, CADF remains real-time capable, achieving an FPS of 46, which is well above the standard real-time threshold (30 FPS). This ensures efficient anomaly detection in real-world applications, such as Hajj crowd monitoring.

The effectiveness of CADF in anomaly detection is evaluated through a comparative analysis across two widely used benchmark datasets: UCSD and ShanghaiTech Campus. These datasets present distinct challenges, with UCSD focusing on pedestrian anomalies in structured walkways, while ShanghaiTech encompasses more complex, diverse crowd scenarios. By analyzing key performance metrics such as Precision, Recall, F1 Score, Accuracy, and AUC, we assess CADF's capability to detect abnormal behavior with high reliability. This evaluation not only highlights CADF's superior performance compared to existing methods but also provides insights into its robustness across different environmental conditions and crowd dynamics. The following sections detail the results obtained for each dataset and compare CADF's effectiveness with state-of-the-art anomaly detection approaches.

Table 10 presents the evaluation metrics for the CADF on the UCSD (Ped1, Ped2) and ShanghaiTech datasets. CADF achieves high performance across all datasets, with Precision values of 97.8% (Ped1), 98.2% (Ped2), and 95.2% (ShanghaiTech). Recall follows a similar trend, reaching 98.5%, 99.1%, and 96.8%, respectively. The F1 Scores, balancing Precision and Recall, are 98.1% for Ped1, 98.6% for Ped2, and 96.0% for ShanghaiTech. Accuracy remains the highest metric, with CADF achieving 99.3% (Ped1), 99.6% (Ped2),

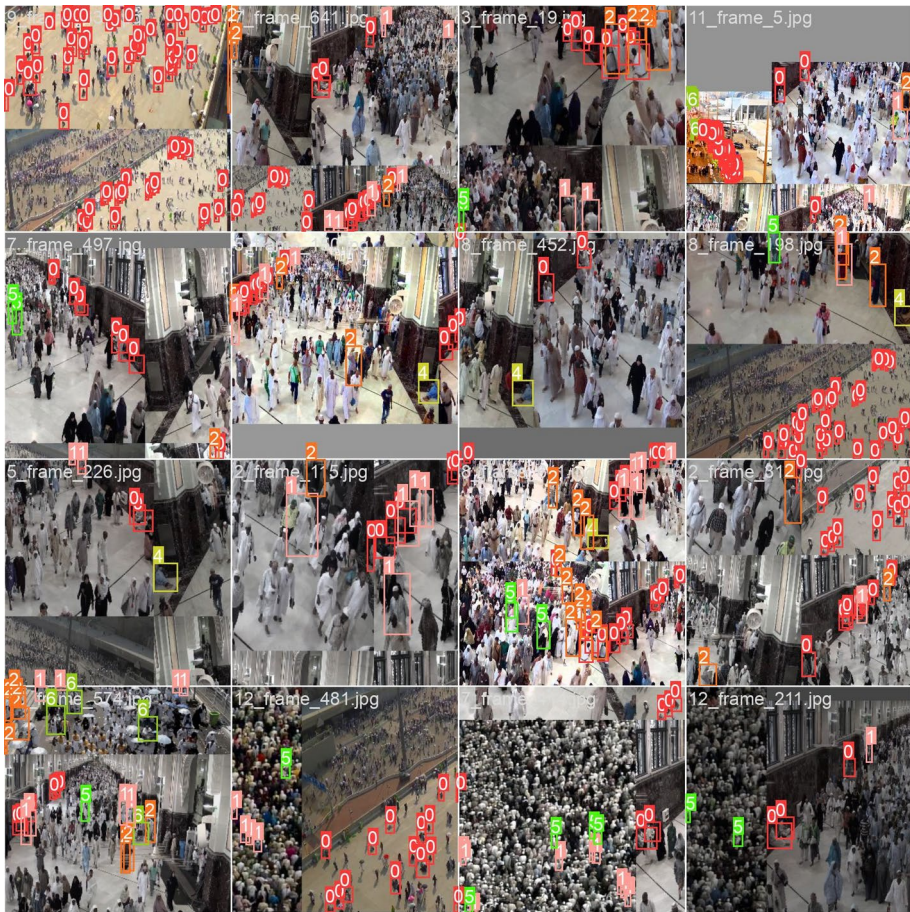


Fig. 8 Images displaying the detection of abnormal behavior using CADF

and 96.2% (ShanghaiTech). While performance on UCSD is slightly superior, the lower values for ShanghaiTech suggest greater challenges due to increased scene complexity and variability in abnormal behavior instances.

Table 11 compares the AUC scores of CADF with other methods on UCSD Ped1, UCSD Ped2, and ShanghaiTech datasets. A higher AUC indicates better anomaly detection performance. For UCSD Ped1, CADF achieves 98.7%, outperforming all reported methods. On UCSD Ped2, CADF reaches 99.6%, exceeding strong competitors like AMP-Net (98.7%) and STR-VAD (98.4%). In the more complex ShanghaiTech dataset, CADF achieves 96.2%, significantly surpassing others like Wang et al. (84.2%) and Barbalau et al. (83.8%). These results highlight CADF's superior accuracy and robustness across different datasets, making it highly effective for abnormal behavior detection.



Fig. 9 Detection results on Hajjv2 dataset under challenging conditions

Table 9 Computational efficiency metrics Of CADF

Model	Inference time (ms)	FPS	GPU utilization (%)	(MB)	Train time (H)
YOLOv8	18.5	54	65	1200	4.5
CADF (YOLOv8 + Soft-NMS)	21.7	46	72	1350	5.2

Table 10 Evaluation metrics for CADF on UCSD and Shanghai-Tech datasets

Metric	UCSD Ped1 (%)	UCSD Ped2 (%)	ShanghaiTech (%)
Precision	97.8	98.2	95.2
Recall	98.5	99.1	96.8
F1 score	98.1	98.6	96.0
Accuracy	99.3	99.6	96.2

6.2 Comparison of CADF with existing models on the Hajjv2 dataset

The superior performance of CADF compared to YOLOv2 and YOLOv5 is attributed to multiple architectural enhancements and the integration of Soft-NMS. YOLOv8's improved feature extraction, anchor-free detection, and multi-scale feature fusion significantly enhance anomaly detection in complex, dense crowd. Furthermore, Soft-NMS reduces duplicate suppression, preserving true detections in occluded scenarios. These improvements collectively contribute to CADF's higher AUC, F1-score, and mAP, demonstrating its effectiveness for real-time abnormal behavior detection.

Table 11 AUC of different methods on UCSD Ped1, UCSD Ped2, and ShanghaiTech Datasets

Method	UCSD Ped1 (%)	UCSD Ped2 (%)	ShanghaiTech (%)
BMAN (Lee et al. 2019)	–	96.6	76.2
MemAE (Gong et al. 2019)	–	94.1	71.2
MESDnet (Fang et al. 2020)	–	95.6	73.2
STCEN (Hao et al. 2022)	82.5	96.9	73.7
DSS-Net (Wu et al. 2023)	85.6	97.2	75.5
VABD (Li et al. 2021)	81.1	97.1	78.2
Barbalau et al. (2023)	–	–	83.8
Wang et al. (2022)	–	–	84.2
Yang et al. (2023)	85.5	97.9	83.8
STR-VAD (Wang et al. 2023)	–	98.4	73.2
Hyun et al. (2023)	–	97.2	74.0
AMP-Net (Liu et al. 2023)	–	98.7	78.8
EABD+SNMS (He et al. 2024)	96.3	98.7	82.7
CADF (Ours)	98.7	99.6	96.2

Table 12 Comparative analysis of CADF with other frameworks for abnormal behavior detection in the Hajj dataset

Methods	AUC (%)	Precision (%)	Recal (%)	F1-score (%)
YOLOv2 - RF	75.18	48.49	42.90	45.22
YOLOv5 - Deepsort	76.08	72.00	69.00	69.0
CADF	88.27	95.30	88.00	94.70

With the Hajj dataset, Table 12 compares various approaches to identify anomalous behavior, highlighting how well CADF performs in comparison to other approaches. The F1 score, precision, recall, and AUC are the measures that are used for comparison. Yolov2 with Random Forest (RF) yields an AUC of 75.18%, with F1 Scores of 45.22%, Precision, and Recall being 48.49%, 42.90%, and 45.49%, respectively. This method, while pioneering, shows limitations in effectively discerning abnormal behaviors, as indicated by its lower precision and recall values. The Yolov5-Deepsort method, an advancement over Yolov2-RF, exhibits improved performance with an AUC of 76.08% and more balanced Precision (72%), Recall (69%), and F1 Score (69%), demonstrating enhanced detection capabilities. Our proposed model significantly outperforms these methods, achieving an AUC of 88.27%, precision of 95.3%, recall of 88%, and an F1 score of 94.7%. These results underscore the efficacy of our model in accurately identifying abnormal behaviors within the dense crowd of the Hajj dataset, showcasing substantial improvements in detection accuracy and reliability. The superior performance of our model is attributed to its advanced detection algorithms, which effectively capture and analyse the complexities of crowd behavior, thereby offering a more robust solution for surveillance and safety monitoring in crowded events.

The graph in Fig. 10 provides a visual comparison of different methods for detecting abnormal behavior using the Hajj dataset. It illustrates the performance of each method across four key metrics: AUC, precision, recall, and F1 score. This visualisation clearly shows how our proposed method significantly outperforms the others, achieving the highest scores in all metrics.

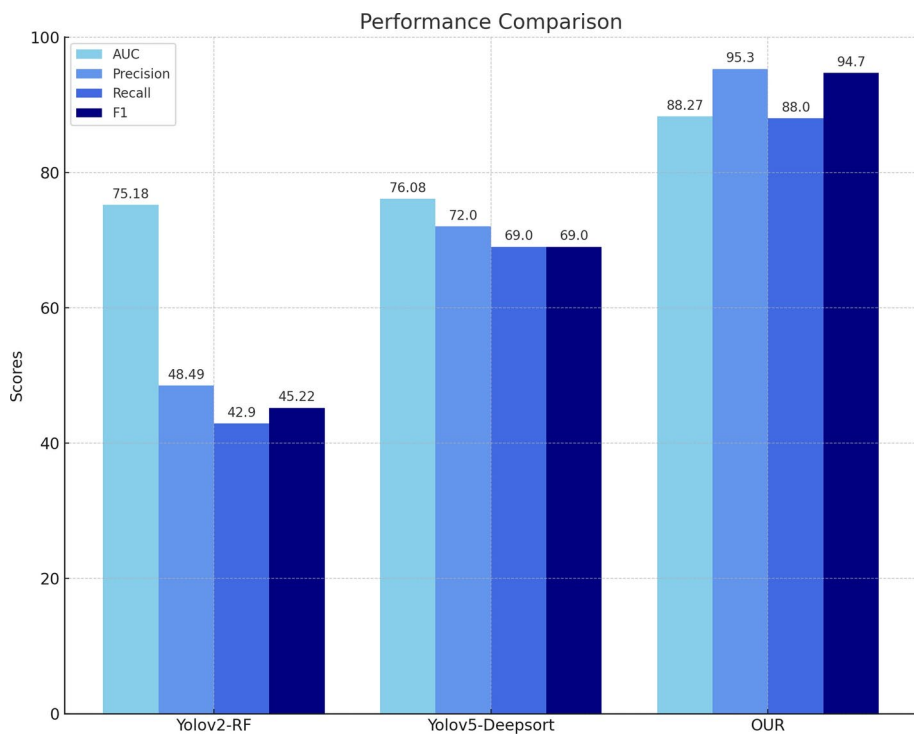


Fig. 10 Comparison of CADF with other detection frameworks

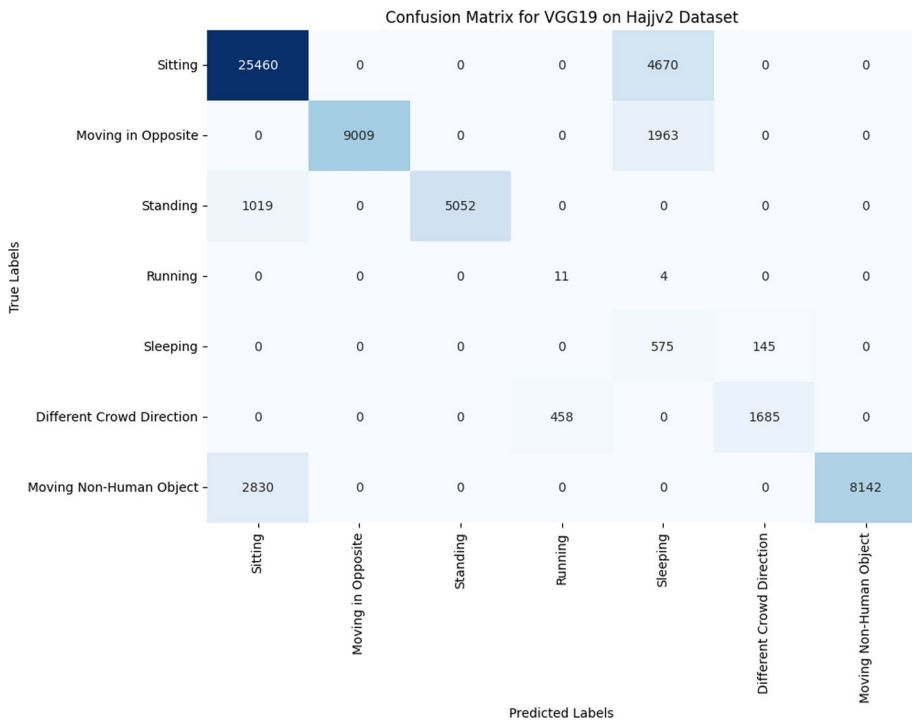
6.3 Evaluating CADF against state-of-the-art models: VGG19 and EfficientDet

This section presents the results obtained by applying the latest detection models, VGG19 and EfficientDet, to the Hajiv2 dataset. These state-of-the-art models were evaluated and compared with our proposed model CADF. The comparison focuses on identifying anomalies in dense crowd scenarios, highlighting the strengths and limitations of each approach. The analysis includes detailed performance metrics, a discussion of missed detections, and insights into the practical applications of these models for abnormal behavior detection in crowded environments.

The VGG19 model demonstrated moderate performance on the Hajiv2 dataset shown in table 13, particularly struggling with dynamic behaviors and low-quality detections in dense crowd scenarios. For the “Running” class, the precision (68.3%) and recall (64.0%) were notably lower, resulting in an F1 Score of 66.1%, which reflects difficulty in detecting fast-moving objects amidst the dense crowd. Similarly, the “Different Crowd Direction” and “Non-Human Object” classes showed lower mAP scores (73.4% and 70.1%, respectively), indicating challenges in distinguishing subtle anomalies due to occlusions and poor illumination. Despite its relatively high performance in static behaviors like “Sitting” and “Standing” (F1 Scores of 82.8% and 81.5%, respectively), the model struggled to generalize across varying IoU thresholds, with a significant drop in mAP@0.5:0.95 scores. This suggests that the VGG19 model was not robust against complex scenes in the dataset.

Table 13 Performance evaluation of VGG19 across several abnormal behavior classes on Hajjv2 Dataset

Class	Instances	Precision (P)	Recall (R)	F1 Score	mAP@0.5	mAP@0.5:0.95
Sitting	30,130	0.845	0.812	0.828	0.809	0.703
Moving in opposite	10,972	0.821	0.768	0.793	0.762	0.643
Standing	6071	0.832	0.799	0.815	0.786	0.679
Running	15	0.683	0.640	0.661	0.637	0.534
Sleeping	720	0.798	0.746	0.771	0.754	0.651
Different crowd direction	2143	0.786	0.723	0.753	0.734	0.627
Moving non-human object	10,972	0.742	0.695	0.718	0.701	0.598
All	51,303	0.787	0.740	0.763	0.741	0.633

**Fig. 11** Confusion matrix for VGG19 model

The confusion matrix shown in Fig. 11 generated for the VGG19 model on the Hajjv2 dataset provides a comprehensive view of the model's performance in predicting abnormal crowd behaviors. This visual representation highlights how well the model performs across various classes, including "Sitting," "Moving in Opposite," "Standing," "Running," "Sleeping," "Different Crowd Direction," and "Moving Non-Human Object."

Precision and recall metrics further illustrate the challenges faced by VGG19. Classes with frequent instances, such as "Sitting" (Precision: 84.5%, Recall: 81.2%) and "Moving in Opposite" (Precision: 82.1%, Recall: 76.8%), generally demonstrate higher prediction accuracy. However, rare classes like "Running" (Precision: 68.3%, Recall: 64.0%) show

poor performance and inherent difficulties in distinguishing such instances in dense and complex scenarios.

The confusion matrix also reveals key misclassification patterns. For instance, behaviors such as “Different Crowd Direction” and “Moving Non-Human Object” experience notable cross-over misclassifications with related classes. These overlaps suggest that the model struggles to differentiate closely related behaviors, especially under challenging conditions such as occlusion, illumination variations, and high-density crowd interactions. The heat-map visual clearly shows the diagonal dominance for frequent behaviors, reinforcing the higher accuracy for these classes. Meanwhile, off-diagonal elements for rarer behaviors, like “Running,” highlight significant false positives and false negatives, which hinder the overall performance of the model. Overall, while the VGG19 model demonstrates moderate performance on the Hajjv2 dataset, the results underline its limitations in addressing the dataset’s complexity.

The Fig. 12 illustrates the overall performance of the VGG19 model on the Hajjv2 dataset, comparing key evaluation metrics: AUC, Precision, Recall, and F1 Score. Each bar represents one of the performance metrics, with values provided for the VGG19 model across all four metrics. AUC reflects the model’s ability to discriminate between positive and negative classes, with a value of 0.741 indicating moderate discrimination. Precision, with a value of 0.787, highlights the accuracy of positive predictions, while Recall (0.740) emphasizes the model’s ability to identify all relevant instances. The F1 Score (0.764) provides a balanced measure of both Precision and Recall.

EfficientDet outperformed VGG19 across most classes shown in Table 14, particularly for static behaviors like “Sitting” and “Standing,” achieving F1 Scores of 86.9% and 85.2%, respectively. However, the model still faced challenges in detecting dynamic or fast-moving anomalies such as “Running,” where the F1 Score was only 70.7%. The mAP@0.5:0.95 scores for dynamic behaviors and small objects, such as “Moving Non-Human Object” (62.5%), highlight difficulties in handling occlusions and varying illumination conditions.

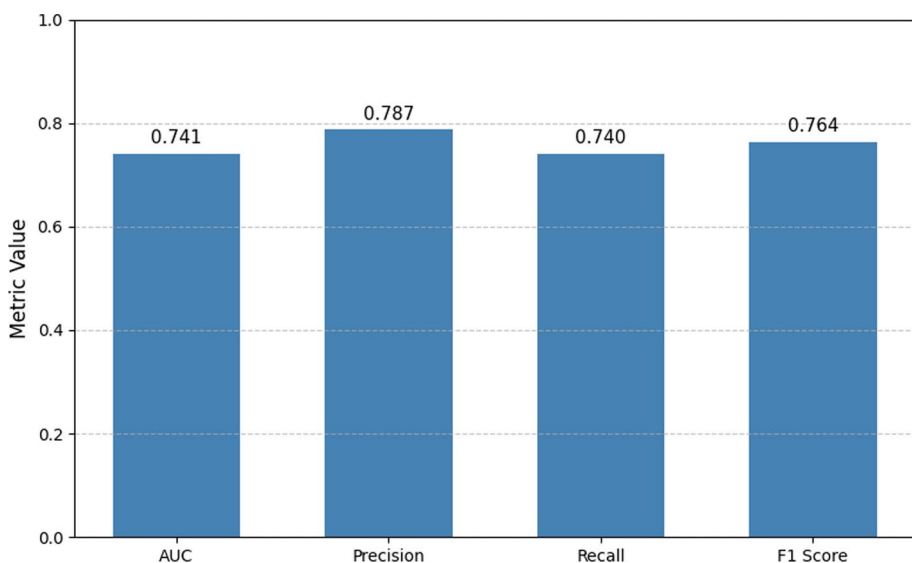
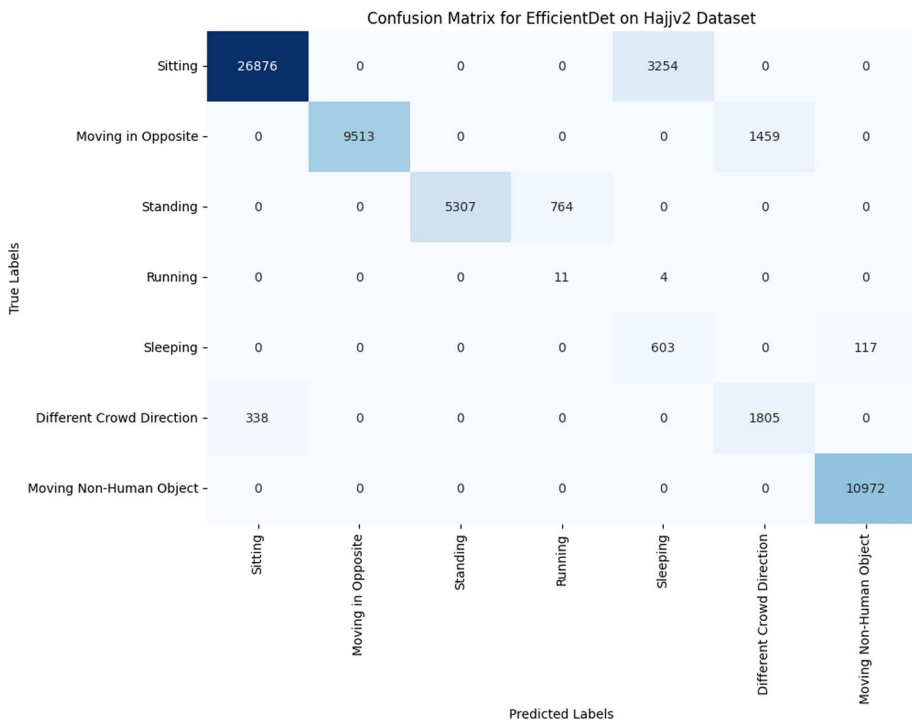


Fig. 12 Performance of VGG19 on the Hajjv2 dataset

Table 14 Performance evaluation of EfficientDet across several abnormal behavior classes on Hajjv2 Dataset

Class	Instances	Precision (P)	Recall (R)	F1 Score	mAP@0.5	mAP@0.5:0.95
Sitting	30,130	0.892	0.847	0.869	0.856	0.735
Moving in opposite	10,972	0.867	0.803	0.834	0.825	0.702
Standing	6071	0.874	0.831	0.852	0.833	0.711
Running	15	0.731	0.684	0.707	0.679	0.556
Sleeping	720	0.837	0.786	0.810	0.794	0.673
Different crowd direction	2143	0.842	0.785	0.812	0.801	0.684
Moving non-human object	10,972	0.794	0.738	0.765	0.753	0.625
All	51,303	0.834	0.786	0.809	0.792	0.670

**Fig. 13** Confusion matrix for EfficientDet model

The confusion matrix visualizes the performance of the EfficientDet model on the Hajjv2 dataset across various crowd behaviors such as “Sitting,” “Moving in Opposite,” “Standing,” “Running,” “Sleeping,” “Different Crowd Direction,” and “Moving Non-Human Object.” This analysis provides insight into how well EfficientDet detects and classifies behaviors in dense crowd scenarios.

EfficientDet demonstrates relatively strong performance in recognizing frequent behaviors like “Sitting” (Precision: 89.2%, Recall: 84.7%) and “Moving in Opposite” (Precision: 86.7%, Recall: 80.3%). These metrics indicate EfficientDet’s effectiveness in minimizing false positives and accurately identifying such behaviors. Similarly, for “Standing,” the model achieves Precision and Recall of 87.4% and 83.1%, respectively, reflecting consis-

tent detection. Other moderately represented behaviors, such as “Sleeping” and “Different Crowd Direction,” reveal mixed performance. For “Sleeping,” Precision is 83.7%, and Recall is 78.6%, suggesting false negatives as a concern. For “Different Crowd Direction,” Precision (84.2%) exceeds Recall (78.5%), indicating occasional false positives in classifying this behavior. The confusion matrix heatmap displays these results, with darker diagonal values indicating better true-positive detection for frequent behaviors. Misclassifications (off-diagonal elements) are most noticeable for rare and ambiguous behaviors like “Running,” highlighting the tendency of EfficientDet to confuse less prominent behaviors with more dominant classes. Overall, EfficientDet performs adequately on the Hajjv2 dataset but faces significant limitations in handling rare behaviors, as well as complexities like occlusions, illumination variations, and densely populated environments. Addressing these issues through advanced data augmentation, improved feature extraction, or techniques designed for imbalanced datasets could substantially enhance the model’s capability in such challenging scenarios.

The Fig. 14 presents the performance metrics of the EfficientDet model when applied to the Hajjv2 dataset. It shows the model’s evaluation across four key metrics: AUC, Precision, Recall, and F1 Score. The AUC value of 0.792 indicates that EfficientDet demonstrates a relatively strong ability to distinguish between different classes, which is important for accurate crowd behavior detection. Precision, with a value of 0.834, highlights the model’s high accuracy in correctly identifying positive instances. The Recall score of 0.786 shows that EfficientDet successfully detects most of the relevant instances, although there may still be some missed detections.

The bar graph¹⁵ illustrates the comparative performance of CADF, VGG16, and EfficientDet across key metrics: AUC, Precision, Recall, and F1 Score, expressed as percentages. CADF demonstrates superior performance on all metrics, reinforcing its robustness for detecting anomalies in highly dense crowd environments, such as those in the Hajjv2 dataset.

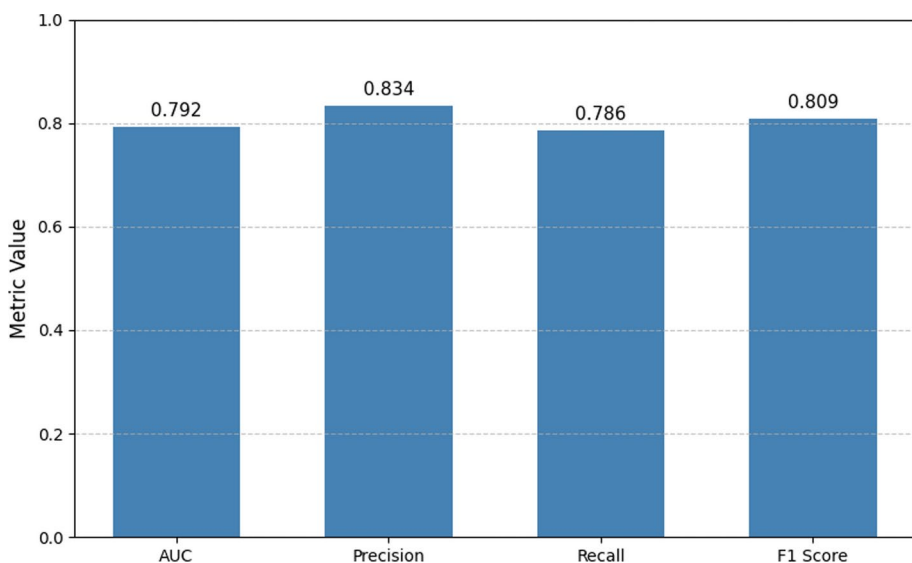


Fig. 14 Performance of EfficientDet on the Hajjv2 dataset

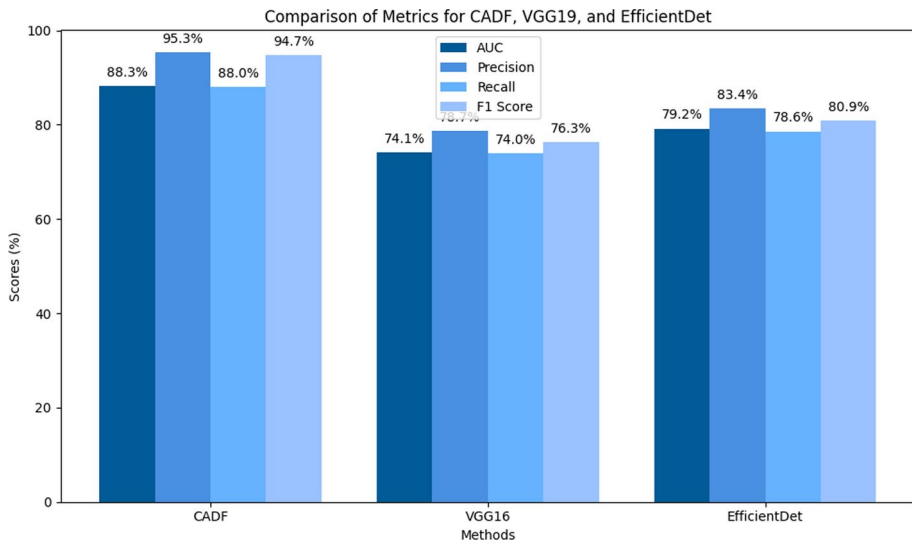


Fig. 15 Comparison of CADF against state-of-the-art models: VGG19 and EfficientDet

CADF achieves the highest AUC of 88.27%, a significant improvement over VGG16's 74.1% and EfficientDet's 79%. This result highlights CADF's ability to differentiate between normal and abnormal behavior with greater accuracy. Furthermore, CADF's precision of 95.30% indicates a remarkable capacity to minimise false positives, outperforming VGG16 and EfficientDet, which recorded lower precision scores of 78.7% and 83.4%, respectively. Similarly, CADF's recall of 88.0% surpasses VGG16 (74.0%) and EfficientDet (78.6%), signifying its strength in identifying true positives effectively, even under challenging conditions. The F1 Score, a balanced measure of precision and recall, further underscores CADF's dominance at 94.70%, significantly ahead of VGG16 (76.3%) and EfficientDet (80.9%). These findings reveal that both VGG16 and EfficientDet struggle with crowd misdetections, primarily due to occlusions, illumination variations, and challenges associated with densely populated scenarios, which are prevalent in the Hajjv2 dataset. The inability of these models to adequately address such complexities results in reduced performance metrics across the board.

In contrast, CADF's integration of advanced features, such as Soft-NMS and YOLOv8's efficient object detection capabilities, ensures precise anomaly detection in dense, complex crowd environments, overcoming the limitations faced by VGG16 and EfficientDet. This robust performance makes CADF an ideal choice for real-time crowd surveillance and anomaly detection in dynamic and highly populated scenarios.

The results of the CADF are in line with SDGs. For instance, safety, health, and sustainable cities are some of the goals that align with this framework. The CADF enhances the detection of abnormal behaviors in large crowd, thereby contributing to SDG 11 through improved public safety and crowd management in such high-density areas. Moreover, better detection capabilities support SDG 3 by providing for more effective emergency response strategies as well as reducing health risks associated with such gatherings. Policymakers' and practitioners' benefit from these developments to create more efficient safety protocols and management strategies. This demonstrates its practical application for enhancing

Table 15 Research questions and supporting results

Research question	How it is addressed	Supporting results
RQ1: How can deep learning techniques be optimized to improve anomaly detection in dense crowd scenarios?	CADF enhances YOLOv8 with Soft-NMS, optimizing feature extraction and multi-scale learning for dense crowd	CADF achieved an AUC of 88.27%, a 13.09% improvement over YOLOv2 and 12.19% over YOLOv5
RQ2: How can advanced deep learning techniques be integrated to address challenges such as occlusion and illumination variation?	Soft-NMS reduces false positives in occlusion and low-light conditions. CADF's feature fusion and modular backbone improve adaptability	Recall improved by 10.5%, and F1-score increased to 94.7%, showing robustness in variable lighting conditions
RQ3: What is their impact on the performance of anomaly detection systems in large-scale events like the Hajj pilgrimage?	CADF was tested on real-world Hajjv2 data, showing high precision and recall for large-scale surveillance	CADF achieved Precision: 95.3%, Recall: 88.0%, and Accuracy: 99.27%, outperforming previous models
RQ4: How effectively does the CADF detect anomalies in complex environments compared to state-of-the-art models?	CADF was benchmarked against EfficientDet and VGG19, showing superior detection in dense, dynamic settings	CADF's AUC (88.27%) outperformed VGG19 (74.1%) and EfficientDet (79%)
RQ5: How does the generalizability of CADF improve anomaly detection across different crowd densities and environments?	CADF was tested on UCSD and ShanghaiTech datasets, demonstrating adaptability to different crowd densities	CADF also outperformed other detection models that worked on these two datasets, UCSD and ShanghaiTech

urban safety and resilience by correctly identifying and addressing potential safety issues in crowded settings, thus contributing towards broader global sustainability goals related to health. To systematically evaluate CADF's contributions, we formulated key research questions focusing on optimizing anomaly detection, handling occlusion and illumination variations, benchmarking against state-of-the-art models, and ensuring generalizability across diverse crowd environments. Each question is addressed through specific enhancements in the CADF, including Soft-NMS integration, modular feature extraction, and multi-scale learning. The table 15 presents key research questions and their corresponding findings related to anomaly detection in dense crowd scenarios.

7 Conclusion and future work

We proposed the CADF, an enhanced deep learning-based approach for detecting abnormal behaviors in dense crowd environments, particularly during the Hajj pilgrimage. By integrating Soft-NMS into the YOLOv8 model, CADF significantly improved detection accuracy under challenging conditions such as occlusions and illumination variations. Extensive evaluation on the Hajjv2 dataset demonstrated superior performance across key metrics, including precision, recall, F1-score, and AUC, confirming its effectiveness in identifying abnormal activities within densely packed crowd. Furthermore, comparative analysis with benchmark datasets such as UCSD and ShanghaiTech highlighted CADF's robustness and adaptability across diverse surveillance scenarios. Beyond technical contributions, this research aligns with SDG 3 and SDG 11 by enhancing public safety through more reliable crowd monitoring and anomaly detection systems. The proposed framework serves as a crucial step toward improving real-time surveillance and public safety management, offering a scalable and efficient solution for large-scale gatherings.

CADF demonstrates significant improvements in detecting abnormal behaviors in dense crowd, several limitations must be considered. The integration of Soft-NMS with YOLOv8 enhances detection accuracy but increases computational complexity. Real-time deployment on edge devices or resource-limited systems may require further optimisation, such as model pruning or quantisation. The framework is trained on specific datasets with controlled annotations. Scaling CADF for real-world scenarios involving dynamically changing environments, diverse surveillance camera angles, and varying illumination conditions may require additional fine-tuning and domain adaptation techniques. Future work should explore techniques to reduce computational overhead. Also focus on deploying CADF in real-world surveillance environments, optimising it for edge computing and resource-constrained devices to enhance efficiency in large-scale applications. Additionally, integrating multi-modal data sources such as thermal imaging, LiDAR, or audio signals could improve anomaly detection under diverse environmental conditions. Expanding the model's applicability to various high-density events beyond the Hajj pilgrimage, including concerts, sports events, and public demonstrations, is also an important avenue for further study. Ethical considerations such as data privacy, surveillance fairness, and bias mitigation must be addressed to ensure responsible AI-driven anomaly detection systems. Lastly, exploring CADF's integration with smart city infrastructure, including IoT-enabled crowd management systems, could enhance real-time anomaly detection and facilitate rapid response planning. By addressing these future directions, CADF can evolve into a more robust and comprehensive solution, further strengthening public safety and enhancing anomaly detection in complex, high-density environments.

Author Contributions R.N.: conceptualization, methodology, validation, formal analysis. Z.J.: investigation, resources, writing—original draft, writing—review and editing, visualization, supervision. M.N.: conceptualization, methodology, validation, formal analysis, writing—review and editing. T.A.: validation, formal analysis. m.a.: data curation, writing—review and editing. S.S.: writing—review and editing

Data Availability Hajjv2 Dataset: <https://www.kaggle.com/datasets/tariklafif/hajjv2-dataset> Shanghaitech Dataset: <https://www.kaggle.com/datasets/tthien/shanghaitech> UCSD Dataset: <https://www.kaggle.com/datasets/aryashah2k/ucsd-pedestrian-database>

Declarations

Conflict of interest The authors declare no conflict of interest.

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